

# Rényi Diffusion Models

Yirong Shen<sup>1</sup> Lu Gan<sup>2</sup> Cong Ling<sup>1</sup>

## Abstract

The choice of training objective is central to diffusion-based generative modeling in terms of both sample quality and distribution coverage. While standard maximum likelihood training provides a principled objective with strong theoretical grounding, empirical studies indicate that previous training objectives in diffusion models often face an inverse correlation between likelihood optimization and perceptual evaluations. We propose the Rényi diffusion model, a unified generative framework that formulates training objectives using Rényi divergence. This yields a generalized score matching objective providing explicit control over the trade-off between sample quality and distribution coverage. Experiments demonstrate improved balance between density estimation and sample generation performances across multiple datasets without modifying model architectures or sampling procedures.

## 1. Introduction

Diffusion models have emerged as a powerful framework in generative modeling (Sohl-Dickstein et al., 2015; Song & Ermon, 2019; Ho et al., 2020; Song et al., 2021c; Peebles & Xie, 2023; Kingma et al., 2021; Song et al., 2021b), achieving state-of-the-art performance across applications including image synthesis (Rombach et al., 2022; Nichol et al., 2022; Ho et al., 2022; Saharia et al., 2023; Li et al., 2026; Gao et al., 2026; Zheng et al., 2026), audio generation (Kong et al., 2021; Chen et al., 2021), and a variety of downstream tasks (Chen et al., 2024a;b; Zeni et al., 2025; Price et al., 2025; Yang et al., 2026; Chen et al., 2025; 2026). A central component of diffusion-based approaches is the design of the training objective (Kingma & Gao, 2023), which governs both the fidelity of density estimation and the perceptual quality of generated samples. While negative

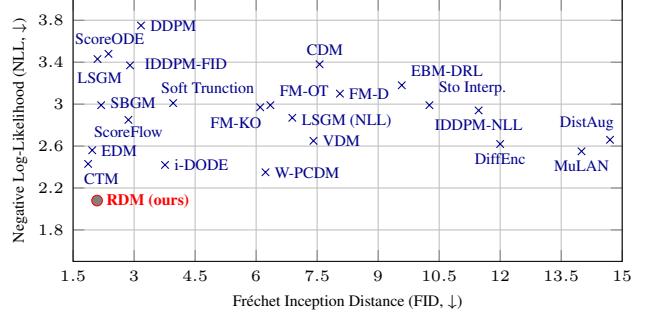


Figure 1. Perceptual quality versus density estimation performance on CIFAR-10. We plot Fréchet Inception Distance (FID, ↓) against Negative Log-Likelihood (NLL) in bits per dimension (BPD, ↓). Each point represents a generative model; baselines are shown in blue and our Rényi Diffusion Model (RDM) in red. RDM achieves a strong balance between sample quality and likelihood. Details of the compared methods are provided in Table 6.

log-likelihood (NLL) provides a principled objective with strong theoretical justification, perceptual metrics such as Fréchet Inception Distance (FID) and Inception Score (IS) are widely used to evaluate sample quality.

Despite substantial progress, empirical evaluations reveal a trade-off between perceptual sample quality and density estimation performance (Kim et al., 2021). As shown in Fig. 1, models achieving strong likelihood performance (low NLL) tend to exhibit weaker perceptual metrics (high FID), while those prioritizing perceptual quality show less competitive likelihood. This pattern persists across diffusion formulations and training objectives, motivating the exploration of alternative objectives that better balance these criteria.

From a probabilistic perspective, most existing diffusion training objectives are derived from the Kullback–Leibler (KL) divergence. KL divergence admits a tractable variational formulation (Wainwright et al., 2008) and is directly connected to the evidence lower bound (ELBO). It gives rise to practical learning objectives such as score matching (Song et al., 2021b), making it a natural and widely adopted choice for likelihood-based training. However, KL divergence is inherently unbounded and asymmetric, and does not define a distance between distributions. As a result, it places strong emphasis on discrepancies in low-probability regions and is highly sensitive to support mismatch, which can bias optimization toward likelihood accuracy while offering limited flexibility in controlling perceptual or geometric similarity

<sup>1</sup>Imperial College London <sup>2</sup>Brunel University of London. Correspondence to: Yirong Shen <ys6922@ic.ac.uk>.

between generated and real data distributions.

Motivated by these limitations, we propose the use of Rényi divergence as a flexible objective for diffusion models. By leveraging its generalized information-theoretic properties, we aim to balance density estimation with perceptual fidelity through a controllable order parameter  $\alpha$ . Our main contributions are summarized as follows:

- We establish a formal connection between Rényi divergence and a generalized *Rényi score matching* objective within the diffusion framework (Theorem 3.1).
- We derive a tighter variational Rényi bound for diffusion models, improving density estimation while preserving perceptual fidelity via the order parameter  $\alpha$  (Section 3.2.2).
- We introduce a weighted Rényi divergence training strategy with importance sampling, and demonstrate on CIFAR-10 and ImageNet that the resulting diffusion models achieve competitive perceptual sample quality (FID and IS) while attaining state-of-the-art density estimation performance, as shown in Fig. 1.

## 2. Background

In this section, we review prior work on diffusion training objectives at a conceptual level; detailed notation and diffusion process definitions are deferred to Appendix B.

### 2.1. Maximum Likelihood Training of Diffusion Models

For density estimation task, NLL has long served as the standard criterion for training and evaluation. Building on variational Bayesian methods (Kingma & Welling, 2014), early diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020) introduced ELBO-based training objectives for tractable NLL optimization. This variational approach was subsequently shown to be KL-consistent and equivalent to score matching (Lyu, 2009; Song et al., 2021b; Kingma et al., 2021; Lu et al., 2022a; Zheng et al., 2023c), establishing NLL as the standard metric for assessing how well diffusion models capture the data distribution (Nichol & Dhariwal, 2021; Kingma et al., 2021; Li et al., 2024a).

However, KL divergence primarily emphasizes statistical fidelity rather than perceptual sample quality. Models trained to minimize NLL thus often exhibit weaker FID scores, reflecting a misalignment between likelihood-based objectives and perceptual metrics.

### 2.2. Perceptual Quality Training of Diffusion Models

Beyond likelihood-based evaluation, perceptual sample quality has become an important criterion in generative

modeling. Metrics such as FID, IS, Kernel Inception Distance (Bińkowski et al., 2018), and Precision/Recall measures (Sajjadi et al., 2018) have been proposed to assess perceptual fidelity, with FID remaining the most widely adopted due to its strong correlation with visual realism.

A key challenge in perceptual quality-driven training is that commonly used metrics such as FID are defined through empirical feature statistics extracted from a fixed Inception network (Szegedy et al., 2016) and do not correspond to directly optimizable loss functions. Existing approaches therefore rely on indirect strategies, such as time-dependent reweighting of the diffusion objective (Choi et al., 2022; Hang et al., 2023; Kingma & Gao, 2023; Hoogetboom et al., 2025; Shi & Titsias, 2025), often combined with adjustments to the noise schedule (Nichol & Dhariwal, 2021; Chen, 2023; Hoogetboom et al., 2023; Zheng et al., 2024a;b; 2025b; Hang et al., 2025) or network parameterization (Salimans & Ho, 2022; Karras et al., 2022; Lipman et al., 2023). While these strategies have improved perceptual metrics such as FID and IS, they often produce reduced likelihood performance (Kim et al., 2021; Li et al., 2024a).

### 2.3. Generalized Objective Functions

Beyond KL-based training, several works have explored more general objective functions for diffusion models to improve robustness, sample quality, or theoretical understanding. From a theoretical perspective, entropic inequalities and optimal transport formulations (Villani et al., 2008; Gentil et al., 2020; Datta & Rouzé, 2017) relate score matching objectives to Wasserstein-type distances (Kwon et al., 2022), providing geometric insights into diffusion dynamics. In parallel, recent  $f$ -divergence-based distillation and refinement methods (Wang et al., 2025; Shi et al., 2024) have demonstrated strong empirical performance in image generation (Tang & Zhou, 2024; Xu et al., 2025; Sun et al., 2025; Wan et al., 2025).

However, these approaches are not designed for likelihood optimization. This motivates the development of alternative divergence-based objectives that offer greater flexibility in balancing likelihood and perceptual fidelity.

## 3. Rényi Diffusion Models

Let  $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N \subset \mathbb{R}^d$  denote a dataset of i.i.d. samples drawn from an unknown data distribution  $p(\mathbf{x})$ . A generative model with parameters  $\theta$  induces a distribution  $q(\hat{\mathbf{x}}; \theta)$ , trained to approximate  $p(\mathbf{x})$  by minimizing a divergence  $D(p(\mathbf{x}) \| q(\hat{\mathbf{x}}; \theta))$ . The choice of divergence critically shapes both the optimization landscape and model behavior (Theis et al., 2016). This work explores Rényi divergence-based diffusion models. A complete summary of notation is provided in Appendix B.1.

### 3.1. Rényi Divergence

Rényi divergence defines a one-parameter family of divergences indexed by  $\alpha \in \mathbb{R}^+$  that generalizes KL divergence:

**Definition 3.1** (Rényi divergence). The parameterized Rényi divergence is defined for two probability distribution  $\rho \ll \nu$  on  $\mathbb{R}^d$  and order  $\alpha \in (0, \infty)$  as:

$$D_\alpha(\rho \parallel \nu) = \begin{cases} \frac{1}{\alpha - 1} \log \left( \int_{\mathbb{R}^d} \left( \frac{\rho}{\nu} \right)^\alpha d\nu \right), & \alpha \neq 1; \\ \int_{\mathbb{R}^d} \frac{\rho}{\nu} \log \left( \frac{\rho}{\nu} \right) d\nu, & \alpha = 1. \end{cases}$$

In the limit  $\alpha \rightarrow 1$ , Rényi divergence recovers the KL divergence that underlies most likelihood-based training in diffusion models. At the other extreme, as  $\alpha \rightarrow 0$ , it reduces to  $D_0(\rho \parallel \nu) = -\log \nu(\{\mathbf{x} : \rho(\mathbf{x}) > 0\})$ , which depends solely on the mass that  $\nu$  assigns to the support of  $\rho$ . A key property is that  $D_\alpha$  is nondecreasing in  $\alpha$ , establishing a continuous interpolation between support-sensitive ( $\alpha \rightarrow 0$ ) and likelihood-aware ( $\alpha \rightarrow 1$ ) regimes for  $\alpha \in (0, 1)$ . This interpolation property makes Rényi divergence particularly attractive for controlling the balance between fit-fidelity mismatch in the generative modeling. Among intermediate orders,  $\alpha = 1/2$  is distinguished by its symmetry and connection to the Hellinger distance, a classical information measure in nonparametric density estimation task (LeCam, 1973; Birgé, 1986; van de Geer, 1993).

While Rényi divergence has been extensively studied in information theory (Harremoës, 2006; Grünwald, 2007; Guo, 2009) and applied in contexts ranging from Bayesian estimation (Haussler & Oppor, 1997; Zhang, 2006; Krishnamurthy et al., 2014; Singh & Póczos, 2014; Li & Turner, 2016) to contrastive learning (Lee & Shin, 2022) and robust optimization (Papernot & Steinke, 2022; Jiang et al., 2023), to our knowledge it has not been explored as a training objective for diffusion models via score matching.

In the following sections, we show how Rényi divergence naturally induces a generalized score-matching objective for diffusion processes. This leads to a principled family of training objectives parameterized by  $\alpha$ , enabling flexible control over the density-estimation versus sample-quality trade-off inherent in diffusion model training.

### 3.2. Rényi-based Objectives for Diffusion Models

#### 3.2.1. RÉNYI SCORE MATCHING

The central objective in score-based diffusion models is to approximate the time-dependent score function  $\nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t)$ , where  $\mathbf{y}_t$  is drawn from the marginal distribution of the diffusion process. A neural network  $\hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta})$  is trained such that  $\hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta}) \approx \nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t)$  for almost all  $\mathbf{y}_t \in \mathbb{R}^d$  and  $t \in [0, 1]$ . Standard score-based approaches

minimize the time-integrated quadratic loss:

$$\ell_{\text{SM}}(w(t); \boldsymbol{\theta}) := \frac{1}{2} \int_0^1 w(t) [\mathcal{J}(p(\mathbf{y}_t) \parallel q(\mathbf{y}_t; \boldsymbol{\theta}))] dt, \quad (1)$$

where  $w : [0, 1] \rightarrow \mathbb{R}^+$  is a positive weighting function and  $\mathcal{J}(\cdot \parallel \cdot)$  denotes the score matching objective

$$\mathcal{J}(p(\cdot) \parallel q(\cdot; \boldsymbol{\theta})) \triangleq \mathbb{E}_{p(\mathbf{y}_t)} [\|\nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t) - \hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2].$$

While generalizations based on various semigroups and entropy measures have been explored (Arnold et al., 2001; 2022; Valero-Toranzo et al., 2017; Shi et al., 2024; Wang et al., 2025; Xu et al., 2025), Rényi-based measures have not been investigated as a training objective for diffusion models. We develop this connection by studying two probability distributions evolving under the heat flow, a setting closely related to mismatched channels in information theory (Guo, 2009; Verdú, 2010; Jiao et al., 2017; Wibisono et al., 2017), where distinct input distributions propagate through an identical channel. This naturally motivates Rényi divergence as a principled measure of discrepancy between the resulting evolutions. All proofs are deferred to the appendix.

**Theorem 3.1.** Let  $W_t : \mathbf{x} \mapsto p(\mathbf{y}_t | \mathbf{x}) = \mathcal{N}(a_t \mathbf{x}, b_t^2 \mathbf{I})$  denote the forward diffusion process at any given time  $t \in [0, 1]$ . Fix  $\alpha \in (0, \infty)$  and set

$$\mathcal{M}_\alpha = \left\{ p(\mathbf{y}_t) \in \mathcal{C}^2(\mathbb{R}^d) : p \ll q, D_\alpha(p \parallel q) < \infty, \nabla_{\mathbf{y}_t}(p/q) \in L^2(q(\mathbf{y}_t; \boldsymbol{\theta}) d\mathbf{y}_t) \right\}. \quad (2)$$

Then for every  $p(\mathbf{y}_t) \in \mathcal{M}_\alpha$  and the map  $b_t^2 \mapsto D_\alpha(p(\mathbf{y}_t) \parallel q(\mathbf{y}_t; \boldsymbol{\theta}))$  is  $\mathcal{C}^1$ , we have

$$\frac{\partial}{\partial b_t^2} D_\alpha(p_t \parallel q_t) = -\frac{1}{2} \frac{\mathcal{J}_\alpha(p(\mathbf{y}_t) \parallel q(\mathbf{y}_t; \boldsymbol{\theta}))}{\int_{\mathbb{R}^d} \left( \frac{p(\mathbf{y}_t)}{q(\mathbf{y}_t; \boldsymbol{\theta})} \right)^\alpha q(\mathbf{y}_t; \boldsymbol{\theta}) d\mathbf{y}_t}, \quad (3)$$

where the variables of distribution in  $D_\alpha$  are omitted for clarity and

$$\mathcal{J}_\alpha(\rho \parallel \nu) = \alpha \left( \int_{\mathbb{R}^d} \rho^\alpha(\mathbf{x}) \nu^{1-\alpha}(\mathbf{x}) \left\| \nabla \log \frac{\rho(\mathbf{x})}{\nu(\mathbf{x})} \right\|_2^2 d\mathbf{x} \right)$$

is the Rényi score matching defined in Definition B.5.

*Sketch of Proof.* The proof relies on the Gaussian smoothing induced by the transition kernel. Let  $p_t$  and  $q_t$  denote the perturbed marginals of the data and model at time  $t$ . These satisfy the heat equation with respect to the variance  $b_t^2$  under the mismatched Gaussian channel setting:

$$\frac{\partial}{\partial b_t^2} p_t = \frac{1}{2} \Delta p_t, \quad \frac{\partial}{\partial b_t^2} q_t = \frac{1}{2} \Delta q_t.$$

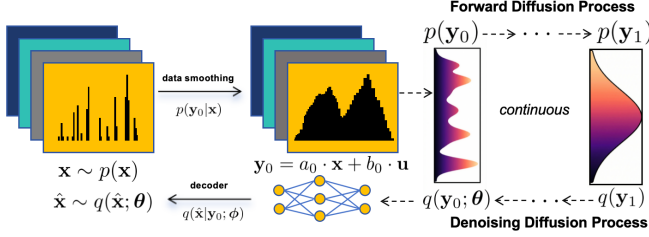


Figure 2. Toy example illustrating our method. We apply a data smoothing process  $p(y_0|x)$  to map discrete data to continuous space and the diffusion process is applied in this space ( $y_0 \rightarrow y_1$ ). Synthesis starts from the Gaussian distribution  $q(y_1)$  and generates samples in continuous space via denoising ( $y_1 \rightarrow y_0$ ). Then, the samples are mapped from continuous to data (pixel) space using a decoder  $q(\hat{x}|y_0; \phi)$ .

Differentiating the Rényi divergence with respect to  $b_t^2$  and applying integration by parts via Green’s identity, we can rewrite the derivative as

$$-\frac{1}{2} \int \left[ \alpha \nabla \left( \left( \frac{p_t}{q_t} \right)^{\alpha-1} \right) \cdot \nabla p_t + (1-\alpha) \nabla \left( \left( \frac{p_t}{q_t} \right)^{\alpha} \right) \cdot \nabla q_t \right] dy.$$

Finally, simplifying the integrand in terms of the log-density gradients yields the key result:

$$\frac{\partial}{\partial b_t^2} D_{\alpha}(p_t || q_t) = -\frac{1}{2} \frac{\mathcal{J}_{\alpha}(p(y_t) || q(y_t; \theta))}{\int_{\mathbb{R}^d} \left( \frac{p(y_t)}{q(y_t; \theta)} \right)^{\alpha} q(y_t; \theta) dy_t}.$$

This establishes the expression in Theorem 3.1 while highlighting the connection between variance evolution and gradient discrepancies under the Rényi divergence.  $\square$

While Theorem 3.1 establishes the theoretical connection to the Rényi score-matching objective, direct optimization remains intractable. Specifically, the model density  $q(y_t; \theta)$  and its score  $\nabla \log q(y_t; \theta)$  are unavailable in closed form during training. Furthermore, the implicit assumption that the model posterior matches the true posterior (i.e.,  $p(y_t|x) \approx q(y_t|\hat{x}; \theta)$ ) does not hold prior to convergence.

To address these issues, we explore this result within a proper *variational inference* framework (Wainwright et al., 2008; Kingma & Welling, 2014; Li & Turner, 2016; Dieng et al., 2017; Wan et al., 2020).

### 3.2.2. VARIATIONAL RÉNYI DIFFUSION OBJECTIVE

Following the variational diffusion framework (Kingma et al., 2021; Vahdat et al., 2021), we introduce an approximate distribution  $q(y_t|\hat{x}; \theta)$  to approximate the true conditional  $p(y_t|x)$ , bypassing the intractable marginals. Under this setting, the synthesis process begins with a given prior  $q(y_1) = \mathcal{N}(\mathbf{0}, \mathbf{I})$  and proceeds via a gradually denoising process to  $q(y_0; \theta)$ , aiming to approximate  $p(x)$ .

To address the fundamental incompatibility between continuous model  $q(y_0; \theta)$  and discrete data  $p(x)$  within the density estimation task, we employ a hybrid architecture. Instead of applying dequantization, we model the representations  $y_0$  continuously, coupled with a discrete decoder  $q(\hat{x}|y_0; \phi)$  mapping back to pixel space; see Fig. 2 for illustration. Formally, the joint hierarchical generative process across  $T + 1$  discretized steps is defined as a Markov process:

$$q(\hat{x}, y; \phi, \theta) := q(\hat{x}|y_0; \phi) q(y_1) \prod_{i=1}^T q(y_{s(i)}|y_{t(i)}; \theta),$$

where  $s(i) = \frac{i-1}{T}$ ,  $t(i) = \frac{i}{T}$  denote the temporal indices.

**Variational Rényi Bound.** To accomplish Rényi score-matching within variational framework, we look to the Variational Rényi Bound (VRB), which offers a tighter bound than the standard ELBO on the likelihood for  $\alpha < 1$  (Li & Turner, 2016):

**Theorem 3.2** (Variational Rényi bound). *For any  $\alpha \in \mathbb{R}^+$ , the negative log-likelihood (NLL) of the model  $q(\hat{x}; \theta)$  admits the upper bound*

$$-\log q(\hat{x}; \theta) \leq \mathcal{L}_{\text{VI}}^{\alpha}(\mathbf{x}). \quad (4)$$

When  $\alpha = 1$ ,  $\mathcal{L}_{\text{VI}}^{\alpha}(\mathbf{x})$  reduces to the standard variational bound derived from the Kullback–Leibler divergence,

$$\mathcal{L}_{\text{VI}}^1(\mathbf{x}) = \mathbb{E}_{p(y_{0:1}|\mathbf{x})} \left[ \log \frac{p(y_{0:1}|\mathbf{x})}{q(\hat{x}, y; \phi, \theta)} \right]. \quad (5)$$

For any  $\alpha \in (0, 1)$ ,  $\mathcal{L}_{\text{VI}}^{\alpha}$  is given by

$$\mathcal{L}_{\text{VI}}^{\alpha}(\mathbf{x}) = \frac{1}{\alpha - 1} \log \mathbb{E}_{p(y_{0:1}|\mathbf{x})} \left[ \left( \frac{p(y_{0:1}|\mathbf{x})}{q(\hat{x}, y; \phi, \theta)} \right)^{\alpha-1} \right],$$

and satisfies

$$\mathcal{L}_{\text{VI}}^{\alpha}(\mathbf{x}) \leq \mathcal{L}_{\text{VI}}^1(\mathbf{x}) \quad (6)$$

implying a tighter NLL bound for  $0 < \alpha < 1$  than the case  $\alpha = 1$ .

*Sketch of Proof.* Starting from the definition of the Rényi divergence and using  $q(y_t|\hat{x}; \theta) = q(y_t, \hat{x}; \theta)/q(\hat{x}; \theta)$ , we obtain

$$\begin{aligned} \log q(\hat{x}; \theta) &= D_{\alpha}(p(y_t|x) || q(y_t|\hat{x}; \theta)) \\ &+ \frac{1}{1-\alpha} \log \mathbb{E}_{p(y_t|x)} \left[ \left( \frac{q(y_t, \hat{x}; \theta)}{p(y_t|x)} \right)^{1-\alpha} \right]. \end{aligned}$$

Since  $D_{\alpha}(\cdot || \cdot) \geq 0$  for  $\alpha > 0$ , this implies the lower bound

$$\log q(\hat{x}; \theta) \geq \frac{1}{1-\alpha} \log \mathbb{E}_{p(y_t|x)} \left[ \left( \frac{q(y_t, \hat{x}; \theta)}{p(y_t|x)} \right)^{1-\alpha} \right].$$

Replacing  $y_t$  by the full path  $y_{0:1}$  and multiplying by  $-1$  gives the desired NLL upper bound.  $\square$

Table 1. Negative log-likelihood (NLL) in bits/dim (BPD), and FID on CIFAR-10 and ImageNet32.

| CIFAR-10                                   |                |                      |                      | ImageNet-32                                |                      |                      |
|--------------------------------------------|----------------|----------------------|----------------------|--------------------------------------------|----------------------|----------------------|
| Method                                     | Type           | NLL ( $\downarrow$ ) | FID ( $\downarrow$ ) | Method                                     | NLL ( $\downarrow$ ) | FID ( $\downarrow$ ) |
| Gated PixelCNN (Van den Oord et al., 2016) | Autoregressive | 3.03                 | —                    | Gated PixelCNN (Van den Oord et al., 2016) | 3.83                 | —                    |
| PixelCNN (Van Den Oord et al., 2016)       | Autoregressive | 3.00                 | —                    | Image Transformer (Parmar et al., 2018)    | 3.77                 | —                    |
| NVAE (Vahdat & Kautz, 2020)                | Autoencoders   | 2.91                 | —                    | NVAE (Vahdat & Kautz, 2020)                | 3.80                 | —                    |
| VAE with IAF (Kingma et al., 2016)         | Autoencoders   | 3.11                 | —                    | PixelSNAIL (Chen et al., 2018)             | 3.80                 | —                    |
| VDVAE (Child, 2021)                        | Autoencoders   | 2.87                 | —                    | VDVAE (Child, 2021)                        | 3.77                 | —                    |
| $\delta$ -VAE (Razavi et al., 2019)        | Autoencoders   | 2.83                 | —                    | $\delta$ -VAE (Razavi et al., 2019)        | 3.77                 | —                    |
| Flow Matching (Lipman et al., 2023)        | Flow           | 2.99                 | 6.35                 | Flow Matching (Lipman et al., 2023)        | 3.53                 | 5.02                 |
| ScoreFlow (Song et al., 2021b)             | Flow           | 2.90                 | 5.40                 | ScoreFlow (Song et al., 2021b)             | 3.82                 | 10.18                |
| i-DODE (Zheng et al., 2023c)               | Flow           | 2.42                 | 3.76                 | i-DODE (Zheng et al., 2023c)               | 3.43                 | 9.09                 |
| ScoreSDE (Song et al., 2021b)              | Diffusion      | 2.99                 | 2.92                 | Reflected DM (Lou & Ermon, 2023)           | 3.74                 | —                    |
| VDM (Kingma et al., 2021)                  | Diffusion      | 2.65                 | 7.60                 | VDM (Kingma et al., 2021)                  | 3.72                 | —                    |
| Soft Truncation (Kim et al., 2021)         | Diffusion      | 2.88                 | 3.45                 | Soft Truncation (Kim et al., 2021)         | 3.85                 | 8.42                 |
| Reflected DM (Lou & Ermon, 2023)           | Diffusion      | 2.68                 | 2.72                 | Reflected DM (Lou & Ermon, 2023)           | 3.74                 | —                    |
| MULAN (Sahoo et al., 2024)                 | Diffusion      | 2.55                 | 17.62                | MULAN (Sahoo et al., 2024)                 | 3.67                 | 13.19                |
| DiffEnc (Nielsen et al., 2024)             | Diffusion      | 2.62                 | 11.20                | DiffEnc (Nielsen et al., 2024)             | 3.46                 | 11.10                |
| W-PCDM (Li et al., 2024a)                  | Diffusion      | 2.35                 | 6.23                 | W-PCDM (Li et al., 2024a)                  | 3.32                 | —                    |
| RDM (Ours)                                 | Diffusion      | <b>2.09</b>          | <b>2.08</b>          | RDM (Ours)                                 | <b>2.94</b>          | <b>3.75</b>          |

**The Rényi Diffusion Bound.** Applying the variational Rényi bound (Theorem 3.2) to the hierarchical diffusion factorization yields an upper bound on the NLL that couples a reconstruction term with a diffusion-trajectory loss. Under the standard diffusion boundary condition at high noise (i.e., the terminal state carries negligible information about  $\mathbf{x}$ ), we can get a simplified bound for  $\alpha \in (0, 1]$  (details of the proof can be found in Appendix D.2):

**Theorem 3.3.** *For any  $\alpha \in (0, 1]$ , assume that the terminal diffusion distribution satisfies  $p(\mathbf{y}_1|\mathbf{x}) \approx \mathcal{N}(\mathbf{0}, \mathbf{I}) \equiv p(\mathbf{y}_1) = q(\mathbf{y}_1)$ . Then the NLL of the model  $q(\hat{\mathbf{x}}; \boldsymbol{\theta})$  admits the following upper bound:*

$$-\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) \leq \mathbb{E}_{\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [-\log q(\hat{\mathbf{x}}|\mathbf{y}_0; \phi)] + \frac{1+\alpha}{4} \mathbb{E}_{t, \epsilon} [(\beta(t) \|\nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t|\mathbf{x}) - \hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2)], \quad (7)$$

where  $\beta(t) = -b_t^2 \dot{\lambda}_t \geq 0$ , ( $\dot{\lambda}_t < 0$ ) and  $\lambda_t := \log \frac{a_t^2}{b_t^2}$  is the log signal-to-noise ratio (log-SNR) at each timestep  $t$ . The neural network  $\hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta})$  is trained such that  $\hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta}) \approx \nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t)$  for almost all  $\mathbf{y}_t \in \mathbb{R}^d$  and  $t \in [0, 1]$ .

*Sketch of Proof.* The VP marginal  $\mathbf{y}_t = a_t \mathbf{x} + b_t \epsilon$  can be realized by a linear Itô diffusion whose log-SNR parametrization satisfies

$$f(t) = -\frac{1}{2}\beta(t), \quad g(t)^2 = \beta(t) = -b_t^2 \dot{\lambda}_t.$$

The reverse-time path measures  $P^x = p(\mathbf{y}_{0:1}|\mathbf{x})$  and  $Q = q(\mathbf{y}_{0:1})$  have the same diffusion coefficient but different score-dependent drifts. By Girsanov’s theorem, their Radon–Nikodym derivative is an exponential martingale involving  $(b_p - b_q)/g(t)$ . Substituting this likelihood ratio into the path-space Rényi divergence gives, assuming matched terminal marginals,  $p(\mathbf{y}_1) = q(\mathbf{y}_1)$ ,

$$\frac{1}{\alpha - 1} \log \mathbb{E}_{P^x} \left[ \exp \left( \frac{\alpha(\alpha - 1)}{2} \int_0^1 \frac{\|b_p - b_q\|^2}{g(t)^2} dt \right) \right].$$

Using the VP conditional score and the  $\epsilon$ -parameterization converts the integrand into the weighted denoising error with weight  $-\dot{\lambda}_t$ . Combining this path-space Rényi term with the reconstruction term yields the NLL upper bound.  $\square$

Specifically, when  $\alpha = 1$ , Eq. (7) reduces to the standard ELBO widely adopted in diffusion models (Ho et al., 2020; Kingma et al., 2021; Vahdat et al., 2021; Song et al., 2021b):

$$-\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) \leq \mathbb{E}_{p(\mathbf{y}_0|\mathbf{x})} [-\log q(\hat{\mathbf{x}}|\mathbf{y}_0; \phi)] + \frac{1}{2} \mathbb{E}_{t, \epsilon} [(\beta(t) \|\nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t|\mathbf{x}) - \hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2)]. \quad (8)$$

Above, we saw that we need to learn a score network  $\hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta})$  that approximate  $\nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t)$  for all noise levels  $\lambda_t$ . Most contemporary diffusion models are optimized towards a noise prediction loss introduced by (Ho et al., 2020). In this case, the score network is typically parameterized through a noise-prediction ( $\epsilon$ -prediction) model:  $\hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta}) = -\hat{\epsilon}(\mathbf{y}_t, t; \boldsymbol{\theta})/b_t^2$ . The equivalent noise-prediction loss is:

$$\mathcal{L}_{\text{diff}}^\epsilon(\mathbf{x}; \boldsymbol{\theta}) = \frac{1+\alpha}{4} \mathbb{E}_{t \sim \mathcal{U}(0,1), \epsilon} \left[ -\frac{d\lambda}{dt} \|\epsilon - \hat{\epsilon}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2 \right].$$

By bridging this theoretical gap, we can seamlessly leverage established architectural primitives and proven hyperparameter configurations (e.g.,  $\mathbf{x}$ -prediction or  $\mathbf{v}$ -prediction strategies) without further structural modification. Derivations and details appear in Appendix E.

From an information-theoretic perspective, Theorem 3.3 shows that the Rényi-divergence-based objective induces a NLL upper bound that is tighter than the standard KL-based ELBO for any order  $\alpha \in (0, 1]$ . Consequently, the likelihood estimation in diffusion models is no longer restricted to the KL-based formulation, and can be decoupled from architectural or weighting choices primarily designed to

Table 2. Negative log-likelihood (NLL) in bits/dim (BPD), Inception Score (IS) and FID on ImageNet 64 and 128.

| ImageNet 64                                     |                      |                      |                   | ImageNet 128                             |                      |                      |                   |
|-------------------------------------------------|----------------------|----------------------|-------------------|------------------------------------------|----------------------|----------------------|-------------------|
| Method                                          | NLL ( $\downarrow$ ) | FID ( $\downarrow$ ) | IS ( $\uparrow$ ) | Method                                   | FID ( $\downarrow$ ) | NLL ( $\downarrow$ ) | IS ( $\uparrow$ ) |
| PixelCNN (Van Den Oord et al., 2016)            | 3.57                 | –                    | –                 | W-PCDM (Li et al., 2024a)                | –                    | 2.64                 | –                 |
| Glow (Kingma & Dhariwal, 2018)                  | 3.81                 | –                    | –                 | RIN (Jabri et al., 2023)                 | 2.75                 | –                    | 144.1             |
| Flow++ (Ho et al., 2019)                        | 3.69                 | –                    | –                 | PaGoDA (Kim et al., 2024b)               | 1.48                 | –                    | 174.3             |
| PixelSNAIL (Chen et al., 2018)                  | 3.52                 | –                    | –                 | Flow Matching (Lipman et al., 2023)      | 20.9                 | 2.90                 | –                 |
| Sparse Transformer (Child, 2019)                | 3.44                 | –                    | –                 | SimDiff (Hooeboom et al., 2023)          | 1.94                 | –                    | 171.9             |
| Sparse Transformer (DistAug) (Jun et al., 2020) | 3.42                 | –                    | –                 | VDM++ (Kingma & Gao, 2023)               | 1.75                 | –                    | 171.1             |
| Very Deep VAE (Child, 2021)                     | 3.52                 | –                    | –                 | DisCo-Diff (Xu et al., 2024)             | 1.73                 | –                    | –                 |
| Improved DDPM (Nichol & Dhariwal, 2021)         | 3.53                 | 2.92                 | –                 | TarFlow (Zhai et al., 2025)              | 2.89                 | –                    | –                 |
| Routing Transformer (Roy et al., 2021)          | 3.43                 | –                    | –                 | Poly-INR (Singh et al., 2023)            | 2.08                 | –                    | 179.6             |
| Flow Matching (Lipman et al., 2023)             | 3.31                 | 14.45                | –                 | BigGAN (Brock et al., 2019)              | 6.02                 | –                    | 145.8             |
| VDM (Kingma et al., 2021)                       | 3.40                 | –                    | –                 | StyleGAN-XL (Sauer et al., 2022)         | 1.81                 | –                    | <b>200.5</b>      |
| W-PCDM (Li et al., 2024a)                       | 2.95                 | –                    | –                 | SPN (Menick & Kalchbrenner, 2019)        | –                    | 3.08                 | –                 |
| SPN (Menick & Kalchbrenner, 2019)               | 3.52                 | –                    | –                 | SiD2, Flop Heavy (Hooeboom et al., 2025) | <b>1.26</b>          | –                    | –                 |
| Gated PixelCNN (Van den Oord et al., 2016)      | 3.48                 | –                    | –                 | CDM2 (Ho & Salimans, 2022)               | 2.43                 | –                    | 158.4             |
| LP-PCDM (Li et al., 2024a)                      | 3.12                 | –                    | –                 | DiffRes101 (Ma et al., 2024)             | 2.19                 | –                    | 187.8             |
| RDM (Ours)                                      | <b>2.77</b>          | <b>1.94</b>          | 59.81             | RDM (Ours)                               | 1.89                 | <b>2.57</b>          | 178.1             |

improve perceptual quality (e.g., FID), without introducing additional looseness in density estimation.

However, existing approaches do not fully exploit this flexibility. In practice, prior works still rely on separately training score networks under FID-favorable and NLL-favorable configurations. A representative observation is that, when adopting the architecture and weighting scheme optimized for sample quality in (Karras et al., 2022), LP-PCDM and W-PCDM (Li et al., 2024a) achieve competitive FID scores of 2.45 and 2.42, respectively, but at the cost of substantially increased BPD (9.55 and 10.31). As a consequence, some approaches tend to specialize toward a compromise (Kim et al., 2021; Esfandiari et al., 2025), resulting in models whose performance across NLL and FID remains competitive, though without consistently excelling in both.

In contrast, we show that the fidelity–likelihood trade-off observed in prior works is not fundamentally caused by architectural constraints, but rather by the choice of likelihood objective. Our analysis reveals that the reliance on the KL-based ELBO necessitates separate FID- and NLL-favorable training regimes, thereby coupling likelihood estimation with architectural and weighting choices. Replacing the ELBO with a Rényi-induced likelihood bound yields a principled objective that preserves better likelihood evaluation while supporting sample-quality improvements, as follows.

To further align with practical FID-favorable diffusion training, we incorporate a time-dependent weighting function  $w(\lambda)$ , which has been shown to play an irreplaceable role in visual quality optimization (Kingma & Gao, 2023; Karras et al., 2022). During model training, we sample time  $t \sim \mathcal{U}(0, 1)$  then compute  $\lambda_t = f_\lambda(t)$  and assign the corresponding importance sampling distribution  $\rho(\lambda) = -\frac{dt}{d\lambda}$ . This allows us to rewrite the final tractable training objective of  $\mathcal{L}_{\text{diff}}^\epsilon(\mathbf{x}; \theta)$  as  $\mathcal{L}_{\text{diff}}^w(\cdot)$ :

$$\mathcal{L}_{\text{diff}}^w(\cdot) = \mathbb{E}_{\lambda \sim \rho(\lambda), \epsilon} \left[ \frac{1 + \alpha w(\lambda)}{4} \frac{1}{\rho(\lambda)} \|\epsilon - \hat{\epsilon}(\mathbf{y}_t, \lambda_t; \theta)\|_2^2 \right].$$

## 4. Implementation

In this work, we employ a variance-preserving forward process:  $\mathbf{y}_t = a_t \mathbf{x} + b_t \epsilon$ ,  $\forall t \in [0, 1]$ , where  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$  is independent of  $\mathbf{x}$  and  $a_t^2 = \text{sigmoid}(\lambda_t)$ ,  $b_t^2 = \text{sigmoid}(-\lambda_t)$  ensuring the constraint  $a_t^2 + b_t^2 = 1$  followed by (Kingma et al., 2021). As established in prior work (Song et al., 2021a), the reverse-time diffusion process is the Gaussian:  $p(\mathbf{y}_s | \mathbf{y}_t, \mathbf{x}) = \mathcal{N}(\mathbf{y}_s | \tilde{\mu}_t(\mathbf{y}_t, \mathbf{x}), \sigma_p^2 \mathbf{I})$ ; see Appendix C.

To approximate  $p(\mathbf{x})$  by discrete denoising process, we must align the model distribution  $q(\mathbf{y}_s | \mathbf{y}_t; \theta)$  with  $p(\mathbf{y}_s | \mathbf{y}_t, \mathbf{x})$  across  $t \in [0, 1]$ . Given the Gaussian structure of  $p(\mathbf{y}_s | \mathbf{y}_t, \mathbf{x})$ , we parameterize the learnable reverse transition  $q$  as a Gaussian  $\mathcal{N}(\mu_q(\mathbf{y}_t, t; \theta), \sigma_q^2 \mathbf{I})$ . Consequently, the optimization problem reduces to matching the moments of these two Gaussians under the Rényi divergence. For two Gaussians  $p = \mathcal{N}(\mu_p, \sigma_p^2 \mathbf{I})$  and  $q = \mathcal{N}(\mu_q, \sigma_q^2 \mathbf{I})$ , the Rényi divergence admits a closed-form expression:

$$D_\alpha(p \| q) = \frac{\alpha \|\mu_p - \mu_q\|_2^2}{2\sigma_\alpha^2} + \frac{d}{2(1-\alpha)} \log \frac{\sigma_\alpha^2}{\sigma_p^{2(1-\alpha)} \sigma_q^{2\alpha}}, \quad (9)$$

where  $\sigma_\alpha^2 = (1-\alpha)\sigma_p^2 + \alpha\sigma_q^2$ .

**Optimal variance.** Minimizing Eq. (9) involves optimizing both the mean  $\mu_q$  and the variance  $\sigma_q^2$ . Here, we demonstrate that the analytic optimization of the Rényi divergence with respect to  $\sigma_q^2$  aligns with the optimal variance solution addressed in (Bao et al., 2022). This allows us to derive the analytic optimal value  $\sigma_\alpha^{*2}$  and establish bounds in terms of information-theoretic quantities.

**Corollary 4.1** (Optimal Variance under Rényi Divergence). *For the diffusion process defined in Appendix C, the variance  $\sigma_\alpha^{*2}$  minimizing (9) is:*

$$\sigma_\alpha^{*2} = \sigma_p^2 + \alpha \left( \frac{b_{t|s}^4}{a_{t|s}^2 b_t^2} \right) \left( 1 - \frac{b_t^2}{d} J(\mathbf{y}_t) \right), \quad (10)$$

Table 3. Sample quality (FID &amp; IS) and number of function evaluations (NFE) on ImageNet 256.

| ImageNet 256                             |             |         |              |
|------------------------------------------|-------------|---------|--------------|
| Method                                   | FID (↓)     | NFE (↓) | IS (↑)       |
| ADM-G (Dhariwal & Nichol, 2021)          | 4.59        | 250     | 186.7        |
| ADM-U (Dhariwal & Nichol, 2021)          | 3.94        | 250     | 215.8        |
| CDM (Ho et al., 2022)                    | 4.88        | 100     | 158.7        |
| BigGAN-deep, max IS (Brock et al., 2019) | 27.0        | 1       | <b>317.0</b> |
| IDDP (Nichol & Dhariwal, 2021)           | 12.3        | 250     | —            |
| RIN (Jabri et al., 2023)                 | 3.42        | 1000    | 182.0        |
| SimDiff (Hooeboom et al., 2023)          | 2.77        | 512     | 211.8        |
| SiD2, Flop Heavy (Hooeboom et al., 2025) | <b>1.38</b> | 512     | —            |
| VDM++ (Kingma & Gao, 2023)               | 2.12        | 250     | 267.7        |
| MDM (Gu et al., 2024)                    | 3.51        | 250     | —            |
| StyleGAN-XL (Sauer et al., 2022)         | 2.30        | 1       | 265.1        |
| SiD2, small (Hooeboom et al., 2025)      | 1.72        | 512     | —            |
| DiT-XL/2 (Peebles & Xie, 2023)           | 2.27        | 250     | 278.2        |
| LDM-4-G (Rombach et al., 2022)           | 3.60        | 250     | 247.6        |
| PaGoDA (Kim et al., 2024b)               | 1.56        | 1       | 259.6        |
| RDM (Ours)                               | 2.24        | 250     | 268.1        |

where  $J(\mathbf{y}_t)$  is the Fisher information (Definition B.2),  $\sigma_p^2 = b_s^2 b_t^2 / b_t^2$  and  $a_s, a_t, b_s, b_t$  are diffusion coefficients.

This reveals that the optimal variance  $\sigma_\alpha^{*2}$  interpolates between the given variance  $\sigma_p^2$  of true reverse transition distribution  $p(\mathbf{y}_s | \mathbf{y}_t, \mathbf{x})$  and a correction term scaled by  $\alpha$ , with the correction depending on the Fisher information of noisy observation. When  $\alpha \rightarrow 0$ , the correction vanishes; when  $\alpha = 1$ , it recovers the situation in (Bao et al., 2022).

**Proposition 4.1.** *Under mild regularity conditions (Appendix D.3.2), the optimal model variance  $\sigma_q^{*2}$  satisfies:*

$$\sigma_p^2 \leq \sigma_q^{*2} \leq \sigma_p^2 + \left( \frac{a_s b_t}{a_t} - \sqrt{b_s^2 - \sigma_p^2} \right)^2 \left( 1 - \frac{b_t^2}{\sigma_e^2(\mathbf{y}_t)} \right), \quad (11)$$

where  $\sigma_e^2(\mathbf{y}_t) = \frac{1}{2\pi e} \exp\left(\frac{2}{d} \mathcal{H}(\mathbf{y}_t)\right)$  is the entropy power.

Unlike prior bounds (Bao et al., 2022) that assume bounded data support, our result only requires the existence of differential entropy, making it applicable to a broader class of distributions. The upper bound connects optimal variance to the effective variance, the variance of a Gaussian with equivalent entropy, providing an information-theoretic characterization (Costa & Cover, 1984).

**Image decoders** Previous density estimation methods rely on *dequantization* when measuring discrete data likelihood with a continuous model (Ho et al., 2019; Song et al., 2021b), which introduces the train-test mismatch (Zheng et al., 2023c). In contrast, we rely on an independent discrete decoder (Salimans et al., 2017; Ho et al., 2020) designed specifically for data with discrete intensity values. To ensure fair comparison, our baselines in the ablation study employ the identical decoder from (Vahdat & Kautz, 2020).



Figure 3. Selected samples from Rényi Diffusion Model trained on ImageNet-512.

### Algorithm 1 Training of Rényi Diffusion Model

**Input:** dataset  $\mathcal{D}$ , weighting  $w(\lambda)$ , Rényi order  $\alpha \in (0, 1]$ , noise schedule  $f_\lambda(t)$ , endpoints of noise schedule  $\lambda_{\max} = f_\lambda(0)$ , learning rate  $\eta$ .

**Parameters:** diffusion model  $\theta$ , decoder  $\phi$

**repeat**

```

 $\mathbf{x} \sim \mathcal{D}, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
 $t \sim \mathcal{U}(0, 1), \quad \lambda_t \leftarrow f_\lambda(t), \quad \rho(\lambda) \leftarrow -1/f'_\lambda(t)$ 
 $a_t^2 \leftarrow \text{sigmoid}(\lambda_t), \quad b_t^2 \leftarrow \text{sigmoid}(-\lambda_t)$ 
 $\mathbf{y}_t \leftarrow a_t \mathbf{x} + b_t \epsilon$ 
 $\mathcal{L}_{\text{diff}}^w(\cdot) \leftarrow \frac{1+\alpha}{4} \frac{w(\lambda)}{\rho(\lambda)} \|\epsilon - \hat{\epsilon}(\mathbf{y}_t, \lambda_t; \theta)\|_2^2$ 
 $a_0^2 \leftarrow \text{sigmoid}(\lambda_{\max}), \quad b_0^2 \leftarrow \text{sigmoid}(-\lambda_{\max})$ 
 $\mathbf{y}_0 \leftarrow a_0 \mathbf{x} + b_0 \epsilon$ 
 $\mathbf{y}_0(\theta) \leftarrow (\mathbf{y}_0 - b_0 \hat{\epsilon}(\mathbf{y}_0, \lambda_0; \theta)) / a_0$ 
 $\mathcal{L}_{\text{Recon}} \leftarrow -\log q(\hat{\mathbf{x}} | \mathbf{y}_0; \phi)$ 
 $\mathcal{L}_{\text{RDM}} \leftarrow \mathcal{L}_{\text{diff}}^w(\cdot) + \mathcal{L}_{\text{Recon}}$ 
 $\{\theta, \phi\} \leftarrow \{\theta, \phi\} - \eta \nabla_{\theta, \phi} \mathcal{L}_{\text{RDM}}$ 
    
```

**until** converged;

## 5. Experimental Results

We demonstrate empirically that by appropriately tuning the order of Rényi divergence, our approach is able to improve the density estimation of existing diffusion models.

### 5.1. NLL and NLL-FID Trade-off

We evaluate the proposed Rényi Diffusion Model (RDM) primarily from the perspective of *density estimation*, measured by negative log-likelihood (NLL), and its trade-off with sample quality. Unless otherwise stated, all methods use identical architectures and training pipelines as their respective baselines.

**CIFAR-10 and ImageNet-32.** Table 1 reports NLL and FID results on CIFAR-10 and ImageNet-32. On CIFAR-10, RDM achieves an NLL of **2.09** bits/dim, improving over the best prior diffusion-based likelihood model (W-PCDM (Li et al., 2024a), 2.35 bits/dim), while simultaneously attaining a strong FID of **2.08**. This places RDM at the top among diffusion-based models and surpasses strong likelihood-

Table 4. Negative log-likelihood (bits/dim) and sample quality (FID and IS scores) on CIFAR-10 and ImageNet  $64 \times 64$ . Abbreviations: “ $\epsilon$ ” for  $\epsilon$ -parametrization; “ $\nu$ ” for  $\nu$ -parametrization; “NLL” for “Negative Log-Likelihood”; “DS.” for “DDPM sampler”; “ES.” for “EDM sampler”; “WL” for “weighted loss”; and “IS” for “importance sampling”. Bold indicates best result in the corresponding column. Blue rows represent models trained with both weighted loss and importance sampling with Fourier feature. We evaluate NLL for our models via variational Rényi bound with order  $\alpha = 1/2$  for both Rényi-trained model (“Rényi”) and KL-trained model (“baseline”).

| Model                          | Noise Schedule | CIFAR-10    |             |             |             |             | ImageNet $64 \times 64$ |              |             |              |             |
|--------------------------------|----------------|-------------|-------------|-------------|-------------|-------------|-------------------------|--------------|-------------|--------------|-------------|
|                                |                | DS.<br>FID↓ | DS.<br>IS↑  | ES.<br>FID↓ | ES.<br>IS↑  | NLL↓        | DS.<br>FID↓             | DS.<br>IS↑   | ES.<br>FID↓ | ES.<br>IS↑   | NLL↓        |
| Baseline ( $\epsilon$ )        | Cosine         | 6.31        | 9.33        | 6.05        | 9.33        | 2.74        | 2.95                    | 45.78        | 2.88        | 48.31        | 3.57        |
| Baseline ( $\epsilon$ ) + WL   | Cosine         | 2.53        | 9.67        | 2.52        | 9.68        | 3.97        | 1.85                    | 53.57        | 2.18        | 56.61        | 4.42        |
| Baseline ( $\epsilon$ )        | Cosine         | 4.56        | 9.78        | 4.23        | 8.88        | 2.56        | 2.88                    | 54.55        | 2.49        | 58.12        | 3.52        |
| Rényi ( $\epsilon$ )           | Cosine         | 6.33        | 9.35        | 6.09        | 8.30        | 2.48        | 2.85                    | 46.17        | 2.89        | 48.21        | 3.29        |
| Rényi ( $\epsilon$ ) + WL      | Cosine         | 2.51        | 9.68        | 2.47        | 9.61        | 2.66        | 1.84                    | 55.43        | 2.09        | 56.76        | 3.61        |
| Rényi ( $\epsilon$ ) + WL + IS | Cosine         | <b>2.27</b> | 9.72        | 2.20        | 9.75        | 2.24        | <b>1.65</b>             | <b>58.14</b> | 1.45        | 61.32        | 3.16        |
| Rényi ( $\epsilon$ ) + WL + IS | Adaptive       | 2.29        | 9.73        | 2.23        | 9.83        | 2.19        | 1.69                    | 57.51        | 1.43        | 60.67        | 3.09        |
| Rényi ( $\epsilon$ )           | Cosine         | 3.37        | 9.76        | 3.22        | 9.19        | 2.13        | 2.20                    | 54.17        | 1.96        | 58.91        | 2.91        |
| Rényi ( $\epsilon$ )           | Adaptive       | 3.42        | <b>9.86</b> | 3.28        | 9.28        | 2.10        | 2.40                    | 55.21        | 1.94        | 59.81        | <b>2.77</b> |
| Baseline ( $\nu$ )             | Adaptive       | 6.27        | 9.43        | 5.83        | 8.56        | 2.66        | 2.31                    | 49.64        | 2.30        | 52.34        | 3.41        |
| Baseline ( $\nu$ ) + WL        | Adaptive       | 2.64        | 9.78        | 2.35        | 8.80        | 3.21        | 1.80                    | 58.12        | 2.11        | 63.63        | 3.95        |
| Baseline ( $\nu$ ) + WL + IS   | Adaptive       | 2.29        | 9.82        | 2.11        | 9.15        | 2.54        | 1.70                    | 57.99        | 1.43        | 65.62        | 3.21        |
| Baseline ( $\nu$ )             | Adaptive       | 4.38        | 9.83        | 3.98        | 9.07        | 2.53        | 2.11                    | 56.91        | 2.07        | 62.77        | 3.12        |
| Rényi ( $\nu$ )                | Adaptive       | 6.31        | 9.41        | 5.82        | 7.56        | 2.27        | 2.29                    | 49.33        | 2.21        | 52.37        | 3.31        |
| Rényi ( $\nu$ ) + WL           | Adaptive       | 2.64        | 9.78        | <b>2.06</b> | 9.05        | 2.78        | 1.78                    | 58.13        | 1.77        | 63.64        | 3.53        |
| Rényi ( $\nu$ ) + WL + IS      | Adaptive       | 2.28        | 9.83        | 2.08        | <b>9.89</b> | 2.09        | 1.67                    | 57.81        | <b>1.42</b> | <b>66.57</b> | 2.98        |
| Rényi ( $\nu$ )                | Adaptive       | 3.41        | 9.84        | 3.17        | 9.77        | <b>2.04</b> | 2.11                    | 54.84        | 2.07        | 63.11        | 2.81        |

Table 5. Negative log-likelihood and sample quality (NLL / FID) on CIFAR-10 and ImageNet.

| Models            | CIFAR-10 |       | ImageNet-32 |       | ImageNet-64 |       |
|-------------------|----------|-------|-------------|-------|-------------|-------|
|                   | FID ↓    | NLL ↓ | FID ↓       | NLL ↓ | FID ↓       | NLL ↓ |
| $\alpha = 0.0001$ | 28.79    | 2.29  | 43.18       | 2.83  | 29.95       | 3.15  |
| $\alpha = 0.1$    | 15.19    | 2.31  | 20.87       | 2.85  | 16.77       | 3.19  |
| $\alpha = 0.3$    | 10.71    | 2.39  | 14.08       | 2.89  | 9.85        | 3.24  |
| $\alpha = 0.5$    | 6.09     | 2.48  | 8.76        | 2.94  | 2.89        | 3.29  |
| $\alpha = 0.7$    | 6.10     | 2.69  | 8.42        | 3.13  | 3.11        | 3.35  |
| $\alpha = 0.9$    | 6.06     | 2.71  | 7.79        | 3.65  | 2.88        | 3.53  |
| $\alpha = 1$      | 6.06     | 2.74  | 7.47        | 3.72  | 2.88        | 3.57  |

based models such as  $\delta$ -VAE (Razavi et al., 2019) (2.83 bits/dim) and i-DODE (Zheng et al., 2023c) (2.42 bits/dim).

On ImageNet-32, where likelihood evaluation is largely restricted to diffusion-based approaches, RDM again attains the lowest reported NLL. In particular, RDM improves upon Flow Matching (Lipman et al., 2023) (3.53 bits/dim) and W-PCDM (Li et al., 2024a) (3.32 bits/dim), while maintaining competitive sample quality. These results suggest that Rényi-based training yields consistent improvements in likelihood without a corresponding severe degradation in perceptual fidelity, unlike the behavior reported in (Sahoo et al., 2024) (3.67 bits/dim with 13.19 FID).

**Higher resolution.** We further evaluate scalability at higher resolution. As shown in Table 2 (right), on ImageNet-128 RDM achieves an NLL of **2.57** bits/dim, outperforming existing diffusion-based likelihood baselines, while retaining a competitive FID of 1.89. This demonstrates that the ben-

## Algorithm 2 Likelihood Evaluation via Rényi Bound

**Input:** Test sample  $\mathbf{x}$ , trained noise-prediction network  $f_\theta$ , trained decoder  $q(\tilde{\mathbf{x}}|\mathbf{y}_0; \phi)$ , Rényi order  $\alpha \in (0, 1]$ , noise schedule  $f_\lambda(t)$ ,  
**Output:** Estimated negative log-likelihood  $\ell(\mathbf{x}; \theta)$   
**Monte Carlo Estimation:**  
 $t \sim \mathcal{U}(0, 1)$ ,  $\lambda_t \leftarrow f_\lambda(t)$ ,  $(a_t^2, b_t^2) \leftarrow \lambda_t$ ,  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
 $\mathbf{y}_t \leftarrow a_t \mathbf{x} + b_t \epsilon$ ,  $\hat{\epsilon} \leftarrow f_\theta(\mathbf{y}_t, \lambda_t)$   
 $\mathcal{L}_{\text{diff}}^\epsilon \leftarrow \frac{1+\alpha}{4} \left( -\frac{d\lambda_t}{dt} \right) \|\epsilon - \hat{\epsilon}(\mathbf{y}_t, \lambda_t)\|_2^2$   
 $\mathcal{L}_{\text{rec}} \leftarrow \mathbb{E}_{p(\mathbf{y}_0|\mathbf{x})} [-\log q(\tilde{\mathbf{x}}|\mathbf{y}_0; \phi)]$   
 Estimate total NLL:  $\ell(\mathbf{x}; \theta) \leftarrow \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{diff}}^\epsilon$   
**return**  $\ell(\mathbf{x}; \theta)$

efits of the Rényi objective extend beyond low-resolution benchmarks, where likelihood evaluation is most common.

## 5.2. Ablation Studies and Effect of the Rényi Order

We next isolate the effect of the Rényi divergence through controlled ablation studies.

Table 4 compares KL- and Rényi-based training under identical architectures and optimization settings. On CIFAR-10, replacing the KL objective with Rényi divergence consistently reduces NLL (e.g., from 2.74 to 2.48 bits/dim under cosine scheduling), while preserving comparable FID. Similar trends are observed on ImageNet-64, confirming that the likelihood gains are not tied to a specific parameterization or noise schedule.

Table 5 studies the effect of the Rényi order  $\alpha$ . Reducing  $\alpha$  from 1 consistently improves likelihood across datasets. In particular,  $\alpha = 1$  recovers the standard KL objective,

Table 6. Comparison between our proposed model and other competitive models in the literature in terms of expected Negative Log Likelihood (NLL) computed as bits per dimension (BPD) and FID on CIFAR-10.

| Model                                        | Type           | NLL Evaluation Method | NLL ( $\downarrow$ ) | FID ( $\downarrow$ ) |
|----------------------------------------------|----------------|-----------------------|----------------------|----------------------|
| EBM-DRL-T6 (Gao et al., 2021)                | Energy-based   | MCMC sampling         | 3.18                 | 9.58                 |
| DistAug (Jun et al., 2020)                   | Autoregressive | Exact evaluation      | 2.66                 | 14.7                 |
| FM-OT (Lipman et al., 2023)                  | Flow           | Exact evaluation      | 2.99                 | 6.35                 |
| FM-D (Lipman et al., 2023)                   | Flow           | Exact evaluation      | 3.10                 | 8.06                 |
| i-DODE (Zheng et al., 2023c)                 | Flow           | Exact evaluation      | 2.42                 | 3.76                 |
| Sto Interp. (Albergo & Vanden-Eijnden, 2023) | Flow           | Exact evaluation      | 2.99                 | 10.27                |
| ScoreFlow (Song et al., 2021b)               | Flow           | Likelihood bound      | 2.90                 | 5.40                 |
| ScoreSDE (Song et al., 2021b)                | Diffusion      | Exact evaluation      | 2.99                 | 2.92                 |
| VDM (Kingma et al., 2021)                    | Diffusion      | Likelihood bound      | 2.65                 | 7.60                 |
| Soft Truncation (Kim et al., 2021)           | Diffusion      | Likelihood bound      | 2.88                 | 3.45                 |
| CTM (Kim et al., 2024a)                      | Diffusion      | Exact evaluation      | 2.43                 | <b>1.87</b>          |
| EDM (Karras et al., 2022)                    | Diffusion      | Exact evaluation      | 2.56                 | 2.01                 |
| LSGM (Vahdat et al., 2021)                   | Diffusion      | Likelihood bound      | 3.43                 | 2.10                 |
| LSGM (NLL) (Vahdat et al., 2021)             | Diffusion      | Likelihood bound      | 2.87                 | 6.89                 |
| IDDPN-NLL (Nichol & Dhariwal, 2021)          | Diffusion      | Likelihood bound      | 2.94                 | 11.47                |
| IDDPN-FID (Nichol & Dhariwal, 2021)          | Diffusion      | Likelihood bound      | 3.37                 | 2.90                 |
| MULAN (Sahoo et al., 2024)                   | Diffusion      | Likelihood bound      | 2.55                 | 17.62                |
| DiffEnc (Nielsen et al., 2024)               | Diffusion      | Likelihood bound      | 2.62                 | 11.20                |
| DDPM (Ho et al., 2020)                       | Diffusion      | Likelihood bound      | 3.75                 | 3.15                 |
| W-PCDM (Li et al., 2024a)                    | Diffusion      | Likelihood bound      | 2.35                 | 6.23                 |
| RDM (Ours)                                   | Diffusion      | Likelihood bound      | <b>2.09</b>          | 2.08                 |

while  $\alpha = \frac{1}{2}$ , closely related to the Hellinger distance, achieves lower NLL than the KL baseline while maintaining competitive FID, with similar gains on ImageNet-32 and -64. This suggests that intermediate Rényi objectives offer a favorable trade-off between likelihood and sample quality.

### 5.3. Training Complexity

Importantly, while RDM does introduce some additional training overhead, it is substantially smaller than that required by variational dequantization based on normalizing flows (e.g., Flow++ (Ho et al., 2019)), as employed in (Song et al., 2021b). Across all experiments, RDM converges within approximately **770k** training iterations, compared to **2M** iterations for i-DODE (Zheng et al., 2023c) and W-PCDM (Li et al., 2024a), and **10M** iterations for VDM (Kingma et al., 2021). This suggests that the tighter variational bound provided by Rényi divergence also leads to favorable optimization dynamics in practice, rather than relying on extended training.

**Sample quality.** While the main text focuses on maximum likelihood training of diffusion models, we report the IS and FID scores of RDM, as shown in Table 3 and Table 7. Since our architectures and training techniques are not explicitly optimized for visual sample quality, the resulting FID is not state-of-the-art but remains competitive, and could be further improved with perceptually oriented algorithms.

**Appendix overview.** The appendix reports additional sample-quality results, including extended FID/IS/NFE comparisons, higher-resolution ImageNet experiments, implementation details, and ablations. It also illustrates NLL-favorable optimization on downstream image compression.

Additional random samples are shown in Appendix K.

## 6. Conclusion

In this work, we present a unified perspective on diffusion-model objectives within the Rényi divergence. We further establish a novel relationship between Rényi divergence and the score matching, thereby providing a generalized insight. Building on this, we introduce a variational formulation that jointly optimizes likelihood fit and perceptual fidelity. Empirically, our method achieves state-of-the-art likelihood estimation on CIFAR-10 and ImageNet while maintaining competitive sample quality, effectively narrowing the long-standing fidelity–likelihood gap within a single objective.

**Limitations** Despite the promising empirical results, our approach has several limitations. First, there is no explicit theoretical guarantee that improvements in likelihood translate into better sample quality. Second, although our method improves the likelihood–quality trade-off, it does not achieve state-of-the-art FID. Third, diffusion models remain computationally expensive at inference time, with sampling considerably slower than one-step methods. Finally, the weighting function is not theoretically characterized, leaving its choice largely empirical.

## Impact Statement

This work advances theory for diffusion models, contributing to foundational mathematical methodology rather than introducing new application-specific capabilities. The associated risks are similar to those of existing generative models, and broader societal considerations remain an important direction for future research and policy.

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## A. Related Works

Likelihood estimation has long constituted a fundamental challenge in generative modeling. While early approaches, building on variational Bayesian methods (Kingma & Welling, 2014), formulated diffusion models by maximizing a variational lower bound on the data log-likelihood (Sohl-Dickstein et al., 2015; Ho et al., 2020), strictly optimizing this bound often yields suboptimal sample quality. To circumvent the intractability of the partition function inherent in maximum likelihood estimation (MLE), subsequent research shifted toward *alternative* objective function schemes, most notably score matching (Song et al., 2021c; Lu et al., 2022a; Zheng et al., 2023c). Score matching serves as a robust alternative to MLE for unnormalized models, effectively bypassing the difficulties associated with explicit Hessian computation (Vincent, 2011; Song et al., 2020). Despite this shift, the theoretical connection between these objectives remains a focal point of study. Recent works (Song et al., 2021b; Kingma et al., 2021; Wibisono et al., 2017; Lyu, 2009; Guo, 2009) have formally bridged the gap, relating the Fisher divergence underlying score matching to the Kullback–Leibler divergence fundamental to MLE, thereby offering a unified perspective on this enduring problem.

Optimizing likelihood in diffusion models is a well-studied problem, with approaches ranging from direct bound estimation (Kingma et al., 2021; Song et al., 2021b; Vahdat et al., 2021) to network parameterization-based optimization (Zheng et al., 2023c). While some studies diverge by learning the forward process (Sahoo et al., 2024; Nielsen et al., 2024; Bartosh et al., 2024; Vahdat et al., 2021), we focus on the standard fixed forward process regime. Here, the fidelity–likelihood gap remains a persistent bottleneck, heavily influenced by the truncation hyperparameter. To address the trade-off between NLL and FID, recent works have explored randomizing the minimum diffusion time  $t_0$ . Strategies include incorporating GAN losses (Kim et al., 2024a) or adopting a Mixture-of-Experts strategy (Esfandiari et al., 2025) that dynamically switches between image-quality and likelihood experts based on the noise level.

Beyond the optimization of the diffusion process itself, recent works have also investigated architectural enhancements inspired by classical signal processing. Techniques such as integrating Fourier features (Kingma et al., 2021) have been employed to better capture high-frequency details, while Wavelet-based approaches have been introduced to facilitate multi-resolution processing within the network (Li et al., 2024a). These modifications aim to strengthen the neural network’s capability to model complex signal characteristics that are often challenging for standard architectures.

## B. Preliminaries and Reviews

### B.1. Notations

In this paper, we are working on the  $d$ -dimensional space  $\mathbb{R}^d$  for some  $d \geq 1$ . We denote the  $\ell_2$ -inner product between vectors  $\mathbf{u} = (u_1, \dots, u_d)$ ,  $\mathbf{v} = (v_1, \dots, v_d) \in \mathbb{R}^d$  as  $\mathbf{u}^\top \mathbf{v} = \langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^d u_i v_i$ .

For a symmetric matrix  $\mathbf{A} \in \mathbb{R}^{d \times d}$ , the notation  $\mathbf{A} \succeq 0$  means  $\mathbf{A}$  is positive semidefinite, i.e.,  $\mathbf{u}^\top \mathbf{A} \mathbf{u} \geq 0$  for all  $\mathbf{u} \in \mathbb{R}^d$ . For symmetric matrices  $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{d \times d}$ , the notation  $\mathbf{A} \succeq \mathbf{B}$  means  $\mathbf{A} - \mathbf{B} \succeq 0$  is positive semidefinite. Throughout, let  $\mathbf{I} \in \mathbb{R}^{d \times d}$  denote the identity matrix.

For a differentiable function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$ , let  $\nabla f(\mathbf{x}) \in \mathbb{R}^d$  denote the gradient vector at  $\mathbf{x} \in \mathbb{R}^d$  of the partial derivatives:  $(\nabla f(\mathbf{x}))_i = \frac{\partial f(\mathbf{x})}{\partial x_i}$ . Let  $\nabla^2 f(\mathbf{x}) \in \mathbb{R}^{d \times d}$  be the Hessian matrix of second partial derivatives:  $(\nabla^2 f(\mathbf{x}))_{i,j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}$ . Let  $\Delta f(\mathbf{x}) = \text{Tr}(\nabla^2 f(\mathbf{x}))$  be the Laplacian. We use  $\mathcal{C}$  to denote all continuous functions, and let  $\mathcal{C}^k$  denote the family of functions with continuous  $k$ -th order derivatives.

For a vector field  $\mathbf{v} : \mathbb{R}^d \rightarrow \mathbb{R}^d$  with  $\mathbf{v}(\mathbf{x}) = (v_1(\mathbf{x}), \dots, v_n(\mathbf{x})) \in \mathbb{R}^d$ , let  $\nabla \mathbf{v} : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$  be the Jacobian matrix of mixed partial derivatives:  $(\nabla \mathbf{v}(\mathbf{x}))_{i,j} = \frac{\partial v_i(\mathbf{x})}{\partial x_j}$ . Let  $\nabla \cdot \mathbf{v} : \mathbb{R}^d \rightarrow \mathbb{R}$  be the divergence of  $\mathbf{v}$ , defined by

$$(\nabla \cdot \mathbf{v})(\mathbf{x}) = \sum_{i=1}^d \frac{\partial v_i(\mathbf{x})}{\partial x_i} = \text{Tr}(\nabla \mathbf{v}(\mathbf{x})). \quad (12)$$

Let  $V_r \subset \mathbb{R}^d$  denote the  $d$ -dimensional ball of radius  $r$  centered at the origin, bounded by the closed, piecewise-smooth, oriented hypersurface  $S_r = \partial V_r$ , i.e., the  $d$ -sphere of radius  $r$ . For any point  $p \in S_r$ , let  $\mathbf{e}_{S_r}(p)$  denote the outward-pointing unit normal vector to  $S_r$ . The oriented surface element is written as

$$ds_r = \|ds_r\| \mathbf{e}_{S_r}(p), \quad (13)$$

where  $\|ds_r\|$  denotes the corresponding scalar surface measure.

More generally, let  $(\mathcal{M}, \mathbf{g})$  be a compact, connected, oriented  $n$ -dimensional Riemannian manifold with (possibly empty) boundary  $\partial\mathcal{M}$ . Throughout this paper, all manifolds are assumed to be smooth, Hausdorff, and second countable; unless otherwise stated, smooth always means  $\mathcal{C}^\infty$ .

We denote by  $T\mathcal{M}$  the tangent bundle of  $\mathcal{M}$ , defined as the disjoint union of the tangent spaces  $T_p\mathcal{M}$  over all points  $p \in \mathcal{M}$ . Note that here  $T\mathcal{M}$  denotes the tangent bundle of  $\mathcal{M}$ , and should not be confused with the scalar variable  $T$  used elsewhere. Each tangent space  $T_p\mathcal{M}$  may be characterized as the space of derivations of the algebra of germs at  $p$  of  $\mathcal{C}^\infty$  functions on  $\mathcal{M}$ , that is, tangent vectors are identified with directional derivatives. The space of smooth sections of the tangent bundle  $T\mathcal{M}$  is denoted by  $\mathcal{T}(\mathcal{M})$ ; equivalently,  $\mathcal{T}(\mathcal{M})$  is the space of smooth vector fields on  $\mathcal{M}$ .

## B.2. Definitions

Let  $\mathcal{P}(\mathbb{R}^d)$  denote the space of probability distributions  $\rho$  over  $\mathbb{R}^d$  which are absolutely continuous with respect to the Lebesgue measure and have a finite second moment  $\mathbb{E}_\rho[\|\mathbf{X}\|^2] < \infty$ . We identify a probability distribution  $\rho \in \mathcal{P}(\mathbb{R}^d)$  with its probability density function with respect to the Lebesgue measure, which we also denote by  $\rho: \mathbb{R}^d \rightarrow \mathbb{R}$ , so  $\rho(x) > 0$  and  $\int_{\mathbb{R}^d} \rho(\mathbf{x}) d\mathbf{x} = 1$ .

We say  $\rho$  is absolutely continuous with respect to another distribution  $\nu$ , denoted by  $\rho \ll \nu$ , if  $\nu(\mathbf{A}) = 0$  implies  $\rho(\mathbf{A}) = 0$  for any  $\mathbf{A} \subseteq \mathbb{R}^d$ ; if  $\rho$  and  $\nu$  both have density functions, then  $\rho \ll \nu$  means  $\nu(\mathbf{x}) = 0$  implies  $\rho(\mathbf{x}) = 0$  for all  $\mathbf{x} \in \mathbb{R}^d$ .

**Definition B.1** (Differential entropy). Let  $\mathcal{H}(\cdot) : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$  be the differential Shannon entropy:

$$\mathcal{H}(\rho) = -\mathbb{E}_\rho[\log \rho] = -\int_{\mathbb{R}^d} \rho(\mathbf{x}) \log \rho(\mathbf{x}) d\mathbf{x}. \quad (14)$$

**Definition B.2** (Fisher information). Let  $J(\cdot) : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$  be the Fisher information:

$$J(\rho) = \mathbb{E}_\rho[\|\nabla \log \rho\|_2^2] = -\mathbb{E}_\rho[\Delta \log \rho], \quad (15)$$

and we define  $J(\rho) = +\infty$  if  $\rho$  does not have a differentiable density. The second equality in the definition of  $J(\rho)$  above follows by integration by parts.

Note that the expression in (15) is a special case (with respect to a translation parameter) which does not involve an explicit parameter as in its most general definition.<sup>1</sup>

**Definition B.3** (Kullback-Leibler divergence). For probability distributions  $\rho \ll \nu$  on  $\mathbb{R}^d$ , the *Kullback-Leibler (KL) divergence* or the *relative entropy* of  $\rho$  with respect to  $\nu$  is defined by:

$$D_{\text{KL}}(\rho\|\nu) = \mathbb{E}_\rho\left[\log \frac{\rho}{\nu}\right] = \int_{\mathbb{R}^d} \rho(\mathbf{x}) \log \frac{\rho(\mathbf{x})}{\nu(\mathbf{x})} d\mathbf{x}. \quad (16)$$

**Definition B.4** (Rényi divergence). The parameterized Rényi divergence is defined for two probability distribution  $\rho \ll \nu$  on  $\mathbb{R}^d$  and order  $\alpha \in (0, \infty)$  as:

$$D_\alpha(\rho\|\nu) = \begin{cases} \frac{1}{\alpha-1} \log \left( \int_{\mathbb{R}^d} \left( \frac{\rho}{\nu} \right)^\alpha d\nu \right), & 0 < \alpha < \infty, \alpha \neq 1; \\ \int_{\mathbb{R}^d} \frac{\rho}{\nu} \log \left( \frac{\rho}{\nu} \right) d\nu, & \alpha = 1. \end{cases} \quad (17)$$

If  $\rho$  is not absolutely continuous with respect to  $\nu$ , simply set  $D_\alpha(\rho\|\nu) = \infty$ . With fixed smooth distributions  $\rho, \nu$  such that  $\rho \ll \nu$ , Rényi divergence is continuous with respect to order  $\alpha$ , in particular,  $D_1(\rho\|\nu) = \lim_{\alpha \rightarrow 1} D_\alpha(\rho\|\nu) = D_{\text{KL}}(\rho\|\nu)$ .

**Definition B.5.** If  $\rho$  and  $\nu$  have differentiable density functions, then the *Rényi score matching* of  $\rho$  with respect to  $\nu$  with

<sup>1</sup>The parameterized Fisher information matrix is defined with respect to a parameter  $\boldsymbol{\theta} \in \boldsymbol{\Theta}$  by  $\mathcal{J}(\rho) \equiv \mathbb{E}_\rho[\nabla_{\boldsymbol{\theta}} \log \rho(\mathbf{x}; \boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}}^\top \log \rho(\mathbf{x}; \boldsymbol{\theta})]$ .

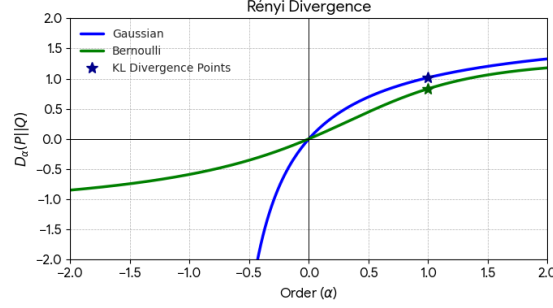


Figure 4. Rényi divergence as a function of its order for specific distributions e.g., Bernoulli and Gaussian.

order  $\alpha \in (0, \infty)$  is defined by:

$$\mathcal{J}_\alpha(\rho||\nu) = \begin{cases} \alpha \left( \int_{\mathbb{R}^d} \rho^\alpha(\mathbf{x}) \nu^{1-\alpha}(\mathbf{x}) \left\| \nabla \log \frac{\rho(\mathbf{x})}{\nu(\mathbf{x})} \right\|_2^2 d\mathbf{x} \right), & 0 < \alpha < \infty, \alpha \neq 1; \\ \int_{\mathbb{R}^d} \rho(\mathbf{x}) \left\| \nabla \log \frac{\rho(\mathbf{x})}{\nu(\mathbf{x})} \right\|_2^2 d\mathbf{x}, & \alpha = 1. \end{cases} \quad (18)$$

### B.3. Properties of Rényi Divergence

Although one can define the Rényi divergence of any order, certain values have wider application than others. Of particular interest are the values 0,  $\frac{1}{2}$ , 1, 2, and  $\infty$ . The values 0, 1, and  $\infty$  are extended orders in the sense that Rényi divergence of these orders cannot be calculated by plugging into (17). Instead, their definitions are determined by continuity in  $\alpha$  (see Figure 4). Specifically, for the extended orders, we have

$$D_0(P||Q) = -\log Q(\{\mathbf{x} : P(\mathbf{x}) > 0\}), \quad (19)$$

which is closely related to absolute continuity and contiguity of the distributions  $P$  and  $Q$ . For order 1, this leads to defining Rényi divergence as the Kullback-Leibler divergence:

$$D_1(P||Q) = D_{\text{KL}}(P||Q) = \text{classical Kullback-Leibler divergence}. \quad (20)$$

For order  $\infty$ , Rényi divergence is the worst-case regret of coding with  $Q$  rather than with  $P$  in the literature on the minimum description length principle in statistics:

$$D_\infty(P||Q) = \log \left( \text{ess sup}_P \frac{p}{q} \right) = \text{worst-case regret}. \quad (21)$$

The Rényi divergence of order  $\infty$  is also related to the separation distance, using to bound the rate of convergence to the stationary distribution for certain Markov chains. Only for  $\alpha = \frac{1}{2}$  is Rényi divergence symmetric in its arguments:

$$D_{1/2}(P||Q) = -2 \log \left( 1 - \frac{\text{Hel}^2(P, Q)}{2} \right), \quad (22)$$

where the squared Hellinger distance  $\text{Hel}^2(P, Q)$  is defined as

$$\text{Hel}^2(P, Q) := \mathbb{E}_Q \left[ \left( 1 - \sqrt{\frac{dP}{dQ}} \right)^2 \right] = \int \left( \sqrt{dP} - \sqrt{dQ} \right)^2 = 2 - 2 \int \sqrt{dP dQ}. \quad (23)$$

Remarkably, the Rényi divergence is nondecreasing in  $\alpha$  over the interval  $[0, \infty]$ , encompassing negative orders, and exhibits continuity in  $\alpha$  on the union of  $[0, 1]$  and the set  $\{\alpha \mid -\infty < D_\alpha < \infty\}$ :

$$\text{Hel}^2(P, Q) \leq D_{1/2}(P||Q) \leq D_1(P||Q) \leq D_2(P||Q) \leq \chi^2(P, Q), \quad (24)$$

where the  $\chi^2$ -divergence is defined as:

$$\chi^2(P, Q) := \mathbb{E}_Q \left[ \left( \frac{dP}{dQ} - 1 \right)^2 \right] = \int \frac{(dP - dQ)^2}{dQ} = \int \frac{dP^2}{dQ} - 1, \quad (25)$$

which imply that

$$D_2(P\|Q) = \log(1 + \chi^2(P, Q)). \quad (26)$$

#### B.4. Evaluation of Diffusion Models

Generative models have many applications and can be evaluated in many ways. For density estimation and related tasks, log-likelihood has been the de-facto standard for training and evaluating generative models (Theis et al., 2016). However, the likelihood of diffusion models is computationally different and difficult. For flow-based diffusion models, likelihood could be computed exactly by numerically solving the probability flow ODE (Lu et al., 2022a; Zheng et al., 2023c) via calling an ODE solver at every optimization step; latent-variable methods optimized the model evidence (Kingma et al., 2021; Sahoo et al., 2024; Nielsen et al., 2024) with the complex integrals; and score-based models highly depend on the closed-form score function of noise.

Beyond likelihood, sample quality has also emerged as an equally important evaluation criterion. A competitive generative model must produce samples that are not only indistinguishable from real data but also capture the full diversity of the underlying distribution, properties that likelihood alone often fails to reflect. To quantitatively measure these aspects, numerous metrics have been proposed, including IS, FID, Kernel Inception Distance (Bińkowski et al., 2018), and Precision/Recall (Sajjadi et al., 2018). Among these metrics, FID continues to be the primary tool for evaluation.

### C. Diffusion Generative Models

Consider a dataset of  $N$  i.i.d. samples  $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$  in  $\mathbb{R}^d$ , where each data point  $\mathbf{x}$  is drawn from the true data distribution  $p(\mathbf{x})$ . In deep generative modeling, a neural network is trained to generate a set of samples  $\hat{\mathbf{x}}$ , which are distributed according to  $q(\hat{\mathbf{x}}; \theta)$  that approximates  $p(\mathbf{x})$ . Several existing methods (Bortoli, 2022) rely on the manifold hypothesis (Chapelle et al., 2006; Narayanan & Mitter, 2010), *i.e.* that real-world high-dimensional data lies in the vicinity of a low-dimensional manifold; see Fig. 6 for the toy demonstration and Appendix I for details.

Score-based diffusion models, a prominent class of generative models via learning the *score function*, can be interpreted through various statistical modeling perspectives (e.g., Song & Ermon, 2019; Song et al., 2021c; Kingma & Gao, 2023; Lu & Song, 2025; Gao et al., 2025; Lu et al., 2022b; Gao et al., 2021; Arriola et al., 2025; Hoogetboom et al., 2022; Zheng et al., 2025a; Karras et al., 2022; Lipman et al., 2023). Despite these differences, they share a common procedure: a forward process gradually perturbs data with noise, and a neural network is trained to reverse this process, generating samples from pure noise.

**Forward Process.** To this end, forward diffusion process defines the joint distribution

$$p(\mathbf{y}) := p(\mathbf{y}_0, \dots, \mathbf{y}_1) = \int p(\mathbf{y}_0, \dots, \mathbf{y}_1 | \mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \quad (27)$$

At any time  $t \in [0, 1]$ , the corrupted observation is defined as

$$\mathbf{y}_t \triangleq a_t \mathbf{x} + b_t \epsilon, \quad \forall t \in [0, 1], \quad (28)$$

where  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$  is independent of  $\mathbf{x}$ . The function  $a_t, b_t^2 \in \mathbb{R}^+$  are smooth, scalar-valued functions with finite derivatives with respect to time  $t$  over the fixed time horizon. The log signal-to-noise ratio (log-SNR) at each timestep is defined by  $\lambda_t \equiv \log(a_t^2/b_t^2)$ . The noise schedule is a strictly decreasing function  $f_\lambda : (0, 1) \rightarrow \mathbb{R}$  that maps time  $t$  to  $\lambda_t$ , *i.e.*,  $\lambda_t = f_\lambda(t)$ . The endpoints of the noise schedule are given by  $\lambda_{\max} := f_\lambda(0)$  and  $\lambda_{\min} := f_\lambda(1)$ . Due to its monotonicity,  $f_\lambda$  is invertible:  $t = f_\lambda^{-1}(\lambda)$ . We sometimes denote the log-SNR as  $\lambda_t$  to emphasize that it is a function of  $t$ .

Then, the forward process is described by a conditional Gaussian distribution:

$$p(\mathbf{y}_t | \mathbf{x}) = \mathcal{N}(a_t \mathbf{x}, b_t^2 \mathbf{I}) = \frac{1}{(2\pi e b_t^2)^{d/2}} \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|_2^2}{2b_t^2} \right), \quad (29)$$

and, by the Markov property,

$$p(\mathbf{y}_t|\mathbf{y}_s) = \mathcal{N}(a_{t|s}\mathbf{y}_s, b_{t|s}^2\mathbf{I}), \quad (30)$$

where

$$\alpha_{t|s} = \frac{a}{a_s}, \quad b_{t|s}^2 = b_t^2 - \frac{a_{t|s}^2}{b_s^2}. \quad (31)$$

Specifically, distribution  $p(\mathbf{y}_0|\mathbf{x})$  is a randomized smoothing process to convolve with discrete data point, which, is *not* necessary Gaussian, making the discrete data distribution smoother. If our goal is to generate approximate samples  $q(\hat{\mathbf{x}}; \boldsymbol{\theta})$ , when  $p(\mathbf{y}_0|\mathbf{x}) = \mathcal{N}(\mathbf{y}_0; a_0\mathbf{x}, b_0^2\mathbf{I})$  and  $b_0^2$  is small, we can use the gradient of  $q(\mathbf{y}_0; \boldsymbol{\theta})$  for denoising. More specifically, given samples  $\mathbf{y}_0$  from  $q(\mathbf{y}_0; \boldsymbol{\theta})$ , we can “denoise” samples via Tweedie formula (Efron, 2011; Alain & Bengio, 2014):

$$\hat{\mathbf{x}} = \frac{1}{a_0} (\mathbf{y}_0 + b_0^2 \nabla_{\mathbf{y}_0} \log q(\mathbf{y}_0; \boldsymbol{\theta})) + o(b_0^2) \quad \text{as } b_0^2 \rightarrow 0. \quad (32)$$

We make use of the small  $o$  notation throughout this paper and assume that the reader is familiar with asymptotic notation. However, this approach does not provide a likelihood estimate and this only works when  $p(\mathbf{y}_0|\mathbf{x})$  is Gaussian. Though alternative method introduces a autoregressive model to invert more general smoothing distributions (beyond Gaussians) and to obtain likelihood estimations (Meng et al., 2021), it is hard to train optimal.

**Reverse Process.** Moreover, given  $s(i) = \frac{i-1}{T}$ ,  $t(i) = \frac{i}{T}$  and  $0 < s < t < 1$ , the diffusion model admits a reverse-time factorization:

$$p(\mathbf{y}|\mathbf{x}) := p(\mathbf{y}_1|\mathbf{x}) \prod_i^T p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x}), \quad (33)$$

where the transitions evolve backward in time from  $t = 1$  to  $t = 0$ , conditioned on the fixed terminal distribution  $p(\mathbf{y}_1|\mathbf{x}) = \mathcal{N}(a_1\mathbf{x}, b_1^2\mathbf{I})$ ; for small enough  $\lambda_{\min}$ ,  $\mathbf{y}_1$  holds almost no information about  $\mathbf{x}$ , such that distribution  $p(\mathbf{y}_1|\mathbf{x})$  is almost identical to  $q(\mathbf{y}_1)$ , which is the starting point of the sampling.

As established in prior work (Ho et al., 2020; Kingma et al., 2021; Song et al., 2021a), the conditional distribution  $p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x})$  is also Gaussian:

$$p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x}) = \mathcal{N}(\mathbf{y}_s|\tilde{\boldsymbol{\mu}}_t(\mathbf{y}_t, \mathbf{x}), \sigma_p^2\mathbf{I}) \quad (34)$$

$$= \mathcal{N}\left(\frac{a_{t|s}b_s^2}{b_t^2}\mathbf{y}_t + \frac{b_{t|s}^2a_s}{b_t^2}\mathbf{x}, \frac{b_s^2b_{t|s}^2}{b_t^2}\mathbf{I}\right). \quad (35)$$

**Generative Model.** As the ground truth  $\mathbf{x}$  is not available during the reverse process, it is approximated by a neural network  $\hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta})$ , parameterised by  $\boldsymbol{\theta}$ . The learned reverse kernel becomes:

$$q(\mathbf{y}_s|\mathbf{y}_t; \boldsymbol{\theta}) = \mathcal{N}\left(\frac{a_{t|s}b_s^2}{b_t^2}\mathbf{y}_t + \frac{b_{t|s}^2a_s}{b_t^2}\hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta}), \sigma_q^2\mathbf{I}\right). \quad (36)$$

In what follows, we define a hierarchical generative model with  $T + 1$  layers of latent variables:  $q(\hat{\mathbf{x}}, \mathbf{y}; \boldsymbol{\phi}, \boldsymbol{\theta}) := q(\hat{\mathbf{x}}, \mathbf{y}_0, \dots, \mathbf{y}_1; \boldsymbol{\phi}, \boldsymbol{\theta})$ :

$$q(\hat{\mathbf{x}}, \mathbf{y}; \boldsymbol{\phi}, \boldsymbol{\theta}) := q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi})q(\mathbf{y}_1) \prod_i^T q(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)}; \boldsymbol{\theta}), \quad (37)$$

$$q(\mathbf{y}_{0:1}; \boldsymbol{\theta}) := q(\mathbf{y}_1) \prod_i^T q(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)}; \boldsymbol{\theta}). \quad (38)$$

In specific, we choose the denoising decoder  $q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi})$  is independent of  $q(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)}; \boldsymbol{\theta})$  and the conditional model distributions as

$$q(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)}; \boldsymbol{\theta}) = p(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)}, \mathbf{x} = \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta})), \quad (39)$$

i.e. the learned reverse kernel  $q(\mathbf{y}_s|\mathbf{y}_t; \boldsymbol{\theta})$ . Finally, we specify the target sampling distribution as  $q(\mathbf{y}_1) = \mathcal{N}(\mathbf{0}, b_1^2\mathbf{I})$ , parameterized by  $b_1^2$ . To this end, with the variance preserving diffusion specification and sufficiently small SNR, the stationary distribution  $p(\mathbf{y}_1|\mathbf{x})$  at the final noise level and prior  $q(\mathbf{y}_1)$  are independent of the reverse transitions.

## D. Proof and Derivation

### D.1. Proof of Theorem 3.1

To prove Theorem 3.1, we first require two lemmas, the proofs of which are provided in the following.

**Lemma D.1** (Heat Equation (Cover, 1999)). *Let an input signal  $\mathbf{x}$  with source density  $p(\mathbf{x})$  pass through the channel in (28), producing an output  $\mathbf{y}_t$  with density  $p(\mathbf{y}_t)$ . Then for every  $\mathbf{y}_t \in \mathbb{R}^d$ ,*

$$\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t) = \frac{1}{2} \Delta_{\mathbf{y}_t} p(\mathbf{y}_t). \quad (40)$$

*Proof.* For brevity, we omit references in the integrals whenever this does not cause ambiguity. Recall that the relationship between the densities of channel input  $\mathbf{x}$  and output  $\mathbf{y}_t$  as follows:

$$p(\mathbf{y}_t) = \int \frac{p(\mathbf{x})}{(2\pi b_t^2)^{\frac{d}{2}}} \exp \left[ -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right] d\mathbf{x}, \quad (41)$$

where  $b_t^2 + a_t^2 = 1$  acts as the scale factor.

**Derivative w.r.t.  $b_t^2$ :** We compute

$$\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t) = \int p(\mathbf{x}) \frac{\partial}{\partial b_t^2} \left[ (2\pi b_t^2)^{-\frac{d}{2}} \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right) \right] d\mathbf{x}. \quad (42)$$

Standard vector calculus yields

$$\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t) = \int p(\mathbf{x}) \frac{\partial}{\partial b_t^2} \left[ (2\pi b_t^2)^{-\frac{d}{2}} \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right) \right] d\mathbf{x} \quad (43)$$

$$= \int p(\mathbf{x}) (2\pi b_t^2)^{-\frac{d}{2}} \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right) \left[ -\frac{d}{2b_t^2} + \frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^4} \right] d\mathbf{x}. \quad (44)$$

**First derivative w.r.t.  $\mathbf{y}_t$ :** Next,

$$\nabla_{\mathbf{y}_t} p(\mathbf{y}_t) = \int p(\mathbf{x}) \nabla_{\mathbf{y}_t} \left[ (2\pi b_t^2)^{-\frac{d}{2}} \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right) \right] d\mathbf{x}. \quad (45)$$

Since  $(2\pi b_t^2)^{-\frac{d}{2}}$  does not depend on  $\mathbf{y}_t$ , only the exponential part matters. We know  $\nabla_{\mathbf{y}_t} \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right) = -\left( \frac{\|\mathbf{y}_t - a_t \mathbf{x}\|}{b_t^2} \right) \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right)$ , so we get

$$\nabla_{\mathbf{y}_t} p(\mathbf{y}_t) = - \int p(\mathbf{x}) (2\pi b_t^2)^{-\frac{d}{2}} \left( \frac{\|\mathbf{y}_t - a_t \mathbf{x}\|}{b_t^2} \right) \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right) d\mathbf{x}. \quad (46)$$

**Laplacian (second derivative) w.r.t.  $\mathbf{y}_t$ :** The Laplacian  $\Delta_{\mathbf{y}_t} p(\mathbf{y}_t)$  is

$$\begin{aligned} \Delta_{\mathbf{y}_t} p(\mathbf{y}_t) &= \text{Tr} \left( \nabla_{\mathbf{y}_t}^\top [\nabla_{\mathbf{y}_t} p(\mathbf{y}_t)] \right) = \nabla_{\mathbf{y}_t} \cdot \nabla_{\mathbf{y}_t} p(\mathbf{y}_t) \\ &= \text{Tr} \left( \int p(\mathbf{x}) (2\pi b_t^2)^{-\frac{d}{2}} \nabla_{\mathbf{y}_t}^\top \left[ -\left( \frac{\|\mathbf{y}_t - a_t \mathbf{x}\|}{b_t^2} \right) \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right) \right] d\mathbf{x} \right). \end{aligned} \quad (47)$$

Applying product rule and simplifying (similarly to the  $b_t^2$  derivative case) gives us

$$\Delta_{\mathbf{y}_t} p(\mathbf{y}_t) = \int p(\mathbf{x}) (2\pi b_t^2)^{-\frac{d}{2}} \left[ -\frac{d}{b_t^2} + \frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{b_t^4} \right] \exp \left( -\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2} \right) d\mathbf{x}. \quad (48)$$

**Relation between  $\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t)$  and  $\Delta_{\mathbf{y}_t} p(\mathbf{y}_t)$**  Comparing this with the expression of  $\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t)$ , we observe that

$$\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t) = \frac{1}{2} \int p(\mathbf{x}) (2\pi b_t^2)^{-\frac{d}{2}} \left[ -\frac{d}{b_t^2} + \frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{b_t^4} \right] \exp\left(-\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2}\right) d\mathbf{x}, \quad (49)$$

while

$$\Delta_{\mathbf{y}_t} p(\mathbf{y}_t) = \int p(\mathbf{x}) (2\pi b_t^2)^{-\frac{d}{2}} \left[ -\frac{d}{b_t^2} + \frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{b_t^4} \right] \exp\left(-\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2}\right) d\mathbf{x}. \quad (50)$$

Hence it follows that

$$\Delta_{\mathbf{y}_t} p(\mathbf{y}_t) = 2 \frac{\partial}{\partial b_t^2} p(\mathbf{y}_t),$$

or equivalently,

$$\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t) = \frac{1}{2} \Delta_{\mathbf{y}_t} p(\mathbf{y}_t). \quad (51)$$

This completes the proof of Lemma D.1.  $\square$

**Lemma D.2** (Vanishing boundary flux). *For an arbitrary input distribution  $p(\mathbf{x})$ , an assumed input distribution  $q(\hat{\mathbf{x}}; \boldsymbol{\theta})$  and a vector Gaussian channel  $p(\mathbf{y}_t | \mathbf{x}) = \mathcal{N}(\alpha_t \mathbf{x}, \sigma_t^2 \mathbf{I})$  in (28), denote the corresponding output densities by  $p(\mathbf{y}_t)$  and  $q(\mathbf{y}_t; \boldsymbol{\theta})$  at time  $t \in [0, 1]$ . For  $r > 0$ , let  $B_r := \{\mathbf{y} \in \mathbb{R}^d : \|\mathbf{y}\| \leq r\}$  be the closed  $d$ -ball and let  $S_r := \partial B_r$  be the  $(d-1)$ -sphere with outward unit normal  $\mathbf{e}_{S_r}(\mathbf{y}) = \mathbf{y}/\|\mathbf{y}\|$ . Assume  $p(\mathbf{y}_t)$  and  $q(\mathbf{y}_t; \boldsymbol{\theta})$  are  $\mathcal{C}^1$  with well-defined  $\nabla_{\mathbf{y}}$  and  $\Delta_{\mathbf{y}}$ , and that  $D_\alpha(p(\mathbf{y}_t) \| q(\mathbf{y}_t; \boldsymbol{\theta})) < \infty$ . Then the boundary integrals*

$$L_1 := \lim_{r \rightarrow \infty} \int_{S_r} \frac{\alpha}{2} \left( \frac{p(\mathbf{y}_t)}{q(\mathbf{y}_t; \boldsymbol{\theta})} \right)^{\alpha-1} \nabla_{\mathbf{y}_t} p(\mathbf{y}_t) \cdot \mathbf{e}_{S_r}(\mathbf{y}_t) dS = 0, \quad (52)$$

$$L_2 := \lim_{r \rightarrow \infty} \int_{S_r} \frac{1-\alpha}{2} \left( \frac{p(\mathbf{y}_t)}{q(\mathbf{y}_t; \boldsymbol{\theta})} \right)^{\alpha} \nabla_{\mathbf{y}_t} q(\mathbf{y}_t; \boldsymbol{\theta}) \cdot \mathbf{e}_{S_r}(\mathbf{y}_t) dS = 0. \quad (53)$$

*Proof.* First, note that the limits  $L_1$  and  $L_2$  obviously exist, as each of these limits can be expressed as the sum of two converging integrals. We prove (52); the argument for (53) is analogous. For brevity, omit the subscript  $t$  and write  $\mathbf{y}$  for  $\mathbf{y}_t$ . Let  $\mathbf{e}_{S_r}(\mathbf{y})$  be unit vector normal to  $S_r$  at the point  $\mathbf{y}$ . Under this notation  $d\mathbf{s}_r = \|d\mathbf{s}_r\| \mathbf{e}_{S_r}(\mathbf{y})$ . We integrate over  $r \geq 0$  the surface integral in (52) and apply Green's identity to find the relations. Set

$$f(r) := \int_{S_r} \frac{\alpha}{2} \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \nabla p(\mathbf{y}) \cdot d\mathbf{s}_r.$$

By the coarea formula (spherical coordinates),

$$\int_0^\infty f(r) dr = \int_{\mathbb{R}^d} \frac{\alpha}{2} \nabla p(\mathbf{y}) \cdot \left( \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \mathbf{e}_{S_r}(\mathbf{y}) \right) d\mathbf{y}. \quad (54)$$

Using the product rule and the divergence theorem on  $B_r$ ,

$$\begin{aligned} \int_{B_r} \frac{\alpha}{2} \nabla p(\mathbf{y}) \cdot \left( \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \mathbf{e}_{S_r}(\mathbf{y}) \right) d\mathbf{y} &= \lim_{r \rightarrow \infty} \frac{\alpha}{2} \int_{S_r} p(\mathbf{y}) \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \mathbf{e}_{S_r}(\mathbf{y}) \cdot d\mathbf{s}_r \\ &\quad - \frac{\alpha}{2} \int_{B_r} p(\mathbf{y}) \nabla \cdot \left( \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \mathbf{e}_{S_r}(\mathbf{y}) \right) d\mathbf{y}. \end{aligned} \quad (55)$$

Letting  $r \rightarrow \infty$  we bound the two terms on the right-hand side.

**First term.** By the coarea formula,

$$\int_r^\infty \frac{\alpha}{2} \int_{S_r} |p(\mathbf{y}) \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1}| dS dr = \int_{\|\mathbf{y}\| > r} \left| \frac{\alpha}{2} p(\mathbf{y}) \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \right| d\mathbf{y} \xrightarrow{r \rightarrow \infty} 0,$$

because  $D_\alpha(p(\mathbf{y})\|q(\mathbf{y})) < \infty$ . Hence  $\int_{S_r} \frac{\alpha}{2} p^\alpha(\mathbf{y}) q^{1-\alpha}(\mathbf{y}) dS \rightarrow 0$  along the full sequence  $r \rightarrow \infty$  (e.g., via a Cesàro argument).

We notice that the absolute value of the divergence in the following term satisfies:

$$|\nabla \cdot (\log q(\mathbf{y}) \mathbf{e}_{S_r}(\mathbf{y}))| = \frac{|\nabla q(\mathbf{y}) \cdot \mathbf{e}_{S_r}(\mathbf{y})|}{q(\mathbf{y})} \leq \frac{\|\nabla q(\mathbf{y})\|}{q(\mathbf{y})}. \quad (56)$$

Hence, we have

$$\begin{aligned} \frac{\|\nabla q(\mathbf{y})\|}{q(\mathbf{y})} &= \left( \sum_{i=1}^n \left[ \int_{\mathbb{R}^d} \frac{q(\mathbf{x})}{q(\mathbf{y})} (2\pi b_t^2)^{-\frac{d}{2}} \left( \frac{\|y_i - a_t x_i\|}{b_t^2} \right) \exp\left(-\frac{\|\mathbf{y}_t - a_t \mathbf{x}\|^2}{2b_t^2}\right) d\mathbf{x} \right]^2 \right)^{\frac{1}{2}} \\ &= \left( \sum_{i=1}^n \left[ \mathbb{E} \left( \frac{Y_i - a_t X_i}{b_t^2} \middle| \mathbf{Y} = \mathbf{y} \right) \right]^2 \right)^{\frac{1}{2}} \\ &\leq \left( \sum_{i=1}^n \left( \mathbb{E} \left( \left( \frac{Y_i - a_t X_i}{b_t^2} \right)^2 \middle| \mathbf{Y} = \mathbf{y} \right) \right) \right)^{\frac{1}{2}} \\ &= \left( \mathbb{E} \left( \left( \frac{\|\mathbf{Y} - a_t \mathbf{X}\|}{b_t^2} \right)^2 \middle| \mathbf{Y} = \mathbf{y} \right) \right)^{\frac{1}{2}}. \end{aligned} \quad (57)$$

Integrating w.r.t.  $p(\mathbf{y}) d\mathbf{y}$  and applying Jensen, we can write the chain of inequalities

$$\begin{aligned} \int_{\mathbb{R}^n} p(\mathbf{y}) \frac{\|\nabla q(\mathbf{y})\|}{q(\mathbf{y})} d\mathbf{y} &\leq \mathbb{E}_p \left\{ \left( \mathbb{E} \left( \left( \frac{\|\mathbf{Y} - a_t \mathbf{X}\|}{b_t^2} \right)^2 \middle| \mathbf{Y} = \mathbf{y} \right) \right)^{\frac{1}{2}} \right\} \\ &\leq \left\{ \mathbb{E}_p \left( \mathbb{E} \left( \left( \frac{\|\mathbf{Y} - a_t \mathbf{X}\|}{b_t^2} \right)^2 \middle| \mathbf{Y} = \mathbf{y} \right) \right) \right\}^{\frac{1}{2}} \\ &= \left[ \mathbb{E}_p \left( \left( \frac{\|\mathbf{N}\|}{b_t^2} \right)^2 \right) \right]^{\frac{1}{2}} \\ &< \infty, \end{aligned} \quad (58)$$

which means that

$$\mathbb{E}_p [\|\nabla \log q(\mathbf{y})\|] \leq \left( \mathbb{E}_p [\|\mathbf{N}\|^2 / b_t^4] \right)^{1/2} < \infty.$$

**Second term.** Similarly, we have

$$\left| \nabla \cdot \left( \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \mathbf{e}_{S_r}(\mathbf{y}) \right) \right| \leq \left\| \nabla \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \right\|. \quad (59)$$

Therefore, we can write

$$\left| \int_{\mathbb{R}^d} p(\mathbf{y}) \nabla \cdot \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \mathbf{e}_{S_r}(\mathbf{y}) d\mathbf{y} \right| \leq \int_{\mathbb{R}^d} p(\mathbf{y}) \left\| \nabla \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \right\| d\mathbf{y} \quad (60)$$

$$= \mathbb{E}_p \left\{ \left\| \nabla \left( \frac{p(\mathbf{y})}{q(\mathbf{y})} \right)^{\alpha-1} \right\| \right\}. \quad (61)$$

Then, we have

$$\begin{aligned}
 \int_{\mathbb{R}^d} p \left\| (\alpha - 1) p^{\alpha-2} q^{1-\alpha} \nabla p + (1 - \alpha) p^{\alpha-1} q^{-\alpha} \nabla q \right\| &= \int_{\mathbb{R}^d} (\alpha - 1) \left\| p^{\alpha-1} q^{1-\alpha} \nabla p + (1 - \alpha) p^{\alpha} q^{-\alpha} \nabla q \right\| \\
 &= \int_{\mathbb{R}^d} (\alpha - 1) \left\| \left( \frac{p}{q} \right)^{\alpha-1} \left( \nabla p - \frac{p}{q} \nabla q \right) \right\| \\
 &= \int_{\mathbb{R}^d} (\alpha - 1) \left\| p \left( \frac{p}{q} \right)^{\alpha-1} \left( \frac{\nabla p}{p} - \frac{\nabla q}{q} \right) \right\| \\
 &= \int_{\mathbb{R}^d} (\alpha - 1) \left\| p \left( \frac{p}{q} \right)^{\alpha-1} (\nabla_{\mathbf{y}} \log p - \nabla_{\mathbf{y}} \log q) \right\|. \quad (62)
 \end{aligned}$$

Due to the assumption that the Rényi score matching is finite, we have (60) is finite. Finally,  $f$  is locally absolutely continuous in  $r$  (by the smoothness of  $p$  and  $q$  and the coarea formula), hence  $\lim_{r \rightarrow \infty} f(r)$  exists. Since  $\int_0^\infty f(r) dr < \infty$ , this limit must equal 0. Thus  $L_1 = 0$ . The proof of  $L_2 = 0$  follows by the same steps.  $\square$

**Theorem 3.1.** Let  $W_t: \mathbf{x} \mapsto p(\mathbf{y}_t|\mathbf{x}) = \mathcal{N}(a_t \mathbf{x}, b_t^2 \mathbf{I})$  denote the forward diffusion process at any given time  $t \in [0, 1]$ . Fix  $\alpha \in (0, \infty)$  and set

$$\mathcal{M}_\alpha = \left\{ p(\mathbf{y}_t) \in \mathcal{C}^2(\mathbb{R}^d) : p \ll q, D_\alpha(p||q) < \infty, \nabla_{\mathbf{y}_t}(p/q) \in L^2(q(\mathbf{y}_t; \boldsymbol{\theta}) d\mathbf{y}_t) \right\}. \quad (63)$$

Then for every  $p(\mathbf{y}_t) \in \mathcal{M}_\alpha$  and the map  $b_t^2 \mapsto D_\alpha(p(\mathbf{y}_t)||q(\mathbf{y}_t; \boldsymbol{\theta}))$  is  $\mathcal{C}^1$ , we have, where the variables of distribution in  $D_\alpha$  are omitted for clarity:

$$\frac{\partial}{\partial b_t^2} D_\alpha(p(\mathbf{y}_t)||q(\mathbf{y}_t; \boldsymbol{\theta})) = -\frac{1}{2} \frac{\mathcal{J}_\alpha(p(\mathbf{y}_t)||q(\mathbf{y}_t; \boldsymbol{\theta}))}{\int_{\mathbb{R}^d} \left( \frac{p(\mathbf{y}_t)}{q(\mathbf{y}_t; \boldsymbol{\theta})} \right)^\alpha q(\mathbf{y}_t; \boldsymbol{\theta}) d\mathbf{y}_t}, \quad (64)$$

where

$$\mathcal{J}_\alpha(\rho||\nu) = \alpha \left( \int_{\mathbb{R}^d} \rho^\alpha(\mathbf{x}) \nu^{1-\alpha}(\mathbf{x}) \left\| \nabla \log \frac{\rho(\mathbf{x})}{\nu(\mathbf{x})} \right\|_2^2 d\mathbf{x} \right)$$

is the Rényi score matching defined in Definition B.5.

**Proof of Theorem 3.1.** Consider the mismatched channel setting where the marginal distributions at time are given by the convolution of the data/model distributions with the transition kernel. Let and denote the marginals:

$$p(\mathbf{y}_t) = \int_{\mathbb{R}^d} p(\mathbf{x}) \mathcal{N}(\mathbf{y}_t|a_t \mathbf{x}, b_t^2 \mathbf{I}) d\mathbf{x}, \quad q(\mathbf{y}_t; \boldsymbol{\theta}) = \int_{\mathbb{R}^d} q(\hat{\mathbf{x}}; \boldsymbol{\theta}) \mathcal{N}(\mathbf{y}_t|a_t \mathbf{x}, b_t^2 \mathbf{I}) d\mathbf{x}. \quad (65)$$

Explicitly,  $p(\mathbf{y}_t)$  is expressed as:

$$p(\mathbf{y}_t) = \frac{1}{(2\pi b_t^2)^{d/2}} \int_{\mathbb{R}^d} p(\mathbf{x}) \exp \left( -\frac{\|\mathbf{y} - a_t \mathbf{x}\|_2^2}{2b_t^2} \right) d\mathbf{x}. \quad (66)$$

Due to the smoothing properties of the Gaussian kernel,  $p(\mathbf{y}_t)$  and  $q(\mathbf{y}_t)$  are smooth functions. Differentiating (66) with respect to the variance parameter  $b_t^2$  under the integral sign (the integrand is continuous and differentiable in  $b_t^2$ ), it is a classical result that both marginals satisfy the heat (diffusion) equation (Lemma D.1):

$$\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t) = \frac{1}{2} \Delta p(\mathbf{y}_t), \quad \frac{\partial}{\partial b_t^2} q(\mathbf{y}_t; \boldsymbol{\theta}) = \frac{1}{2} \Delta q(\mathbf{y}_t; \boldsymbol{\theta}), \quad (67)$$

where  $\Delta$  denotes the Laplacian with respect to  $\mathbf{y}_t$ .

Differentiating the Rényi divergence  $D_\alpha(p||q)$  with respect to  $b_t^2$ , we have:

$$\begin{aligned}\frac{\partial}{\partial b_t^2} D_\alpha(p||q) &= \frac{\partial}{\partial b_t^2} \left[ \frac{1}{\alpha-1} \log \int_{\mathbb{R}^d} p(\mathbf{y}_t)^\alpha q(\mathbf{y}_t; \boldsymbol{\theta})^{1-\alpha} d\mathbf{y}_t \right] \\ &= \frac{1}{\alpha-1} \mathcal{Z}_\alpha^{-1} \frac{\partial}{\partial b_t^2} \int_{\mathbb{R}^d} p(\mathbf{y}_t) \left( \frac{p(\mathbf{y}_t)}{q(\mathbf{y}_t; \boldsymbol{\theta})} \right)^{\alpha-1} d\mathbf{y}_t,\end{aligned}\quad (68)$$

where  $\mathcal{Z}_\alpha = \int p^\alpha(\mathbf{y}_t) q^{1-\alpha}(\mathbf{y}_t; \boldsymbol{\theta}) d\mathbf{y}_t$ . Focusing on the numerator of (68), we exchange differentiation and integration:

$$\begin{aligned}\mathcal{I} &:= \frac{\partial}{\partial b_t^2} \int_{\mathbb{R}^d} p(\mathbf{y}_t) \left( \frac{p(\mathbf{y}_t)}{q(\mathbf{y}_t; \boldsymbol{\theta})} \right)^{\alpha-1} d\mathbf{y}_t \\ &= \int_{\mathbb{R}^d} \left[ \alpha \left( \frac{p(\mathbf{y}_t)}{q(\mathbf{y}_t; \boldsymbol{\theta})} \right)^{\alpha-1} \frac{\partial p(\mathbf{y}_t)}{\partial b_t^2} + (1-\alpha) p(\mathbf{y}_t)^\alpha q(\mathbf{y}_t; \boldsymbol{\theta})^{-\alpha} \frac{\partial q(\mathbf{y}_t; \boldsymbol{\theta})}{\partial b_t^2} \right] d\mathbf{y}_t.\end{aligned}\quad (69)$$

Substituting the heat equations  $\frac{\partial}{\partial b_t^2} p(\mathbf{y}_t) = \frac{1}{2} \Delta p(\mathbf{y}_t)$  and  $\frac{\partial}{\partial b_t^2} q(\mathbf{y}_t; \boldsymbol{\theta}) = \frac{1}{2} \Delta q(\mathbf{y}_t; \boldsymbol{\theta})$ . Denote  $p_t := p(\mathbf{y}_t)$  and  $q_t := q(\mathbf{y}_t; \boldsymbol{\theta})$ :

$$\mathcal{I} = \frac{1}{2} \int_{\mathbb{R}^d} \left[ \alpha \left( \frac{p_t}{q_t} \right)^{\alpha-1} \Delta p_t + (1-\alpha) \left( \frac{p_t}{q_t} \right)^\alpha \Delta q_t \right] d\mathbf{y}_t. \quad (70)$$

We now recall Green's identity: If  $\phi(\mathbf{x})$  and  $\psi(\mathbf{x})$  are twice continuously differentiable functions in  $\mathbb{R}^d$  and  $V$  is any set bounded by a piecewise smooth, closed, oriented surface  $S$  in  $\mathbb{R}^d$ , then

$$\int_V \phi \nabla^2 \psi dV = \int_S \phi \nabla \psi \cdot ds - \int_V \nabla \phi \cdot \nabla \psi dV, \quad (71)$$

where  $\nabla \phi$  denotes the gradient of  $\phi$ ,  $ds$  denotes the elementary area vector, and  $\nabla \phi \cdot ds$  is the inner product of these two vectors. This identity plays the role of integration by parts in  $\mathbb{R}^d$ . To apply Green's identity to (70), we let  $V_r$  be the  $d$ -sphere of radius  $r$  centered at the origin and having surface  $S_r$ . Then we use Green's identity on  $V_r$  and  $S_r$  and take the limit as  $r \rightarrow \infty$ . In Lemma D.2 the surface integral over  $S_r$  is shown to vanish in the limit. Thus, by integration by parts, (70) becomes:

$$\mathcal{I} = -\frac{1}{2} \int_{\mathbb{R}^d} \left[ \alpha \nabla \left( \left( \frac{p_t}{q_t} \right)^{\alpha-1} \right) \cdot \nabla p_t + (1-\alpha) \nabla \left( \left( \frac{p_t}{q_t} \right)^\alpha \right) \cdot \nabla q_t \right] d\mathbf{y}_t. \quad (72)$$

We now simplify the integrand. Let  $r(\mathbf{y}) = p_t/q_t$ . Note that  $\nabla(r^k) = kr^{k-1}\nabla r$ . The integrand in (72) (ignoring the integral sign and factor  $-1/2$ ) is:

$$\alpha(\alpha-1)r^{\alpha-2}(\nabla r \cdot \nabla p_t) + (1-\alpha)\alpha r^{\alpha-1}(\nabla r \cdot \nabla q_t) = \alpha(\alpha-1)r^{\alpha-2}\nabla r \cdot (\nabla p_t - r\nabla q_t). \quad (73)$$

Substituting  $\nabla r = \nabla(p_t/q_t) = \frac{q_t \nabla p_t - p_t \nabla q_t}{q_t^2}$  and noting that  $\nabla p_t - r\nabla q_t = q_t \nabla r$ , the expression becomes:

$$\begin{aligned}\alpha(\alpha-1)r^{\alpha-2}(\nabla r) \cdot (q_t \nabla r) &= \alpha(\alpha-1)r^{\alpha-2}q_t \|\nabla r\|_2^2 \\ &= \alpha(\alpha-1) \left( \frac{p_t}{q_t} \right)^{\alpha-2} q_t \left\| \frac{p_t}{q_t} \nabla \log \frac{p_t}{q_t} \right\|_2^2 \\ &= \alpha(\alpha-1) p_t^\alpha q_t^{1-\alpha} \|\nabla \log p_t - \nabla \log q_t\|_2^2.\end{aligned}\quad (74)$$

Substituting this back into (72) and then into (68), we obtain:

$$\begin{aligned}\frac{\partial}{\partial b_t^2} D_\alpha(p_t||q_t) &= \frac{1}{\alpha-1} \frac{1}{\mathcal{Z}_\alpha} \left( -\frac{1}{2} \int \alpha(\alpha-1) p_t^\alpha q_t^{1-\alpha} \|\nabla \log p_t - \nabla \log q_t\|_2^2 d\mathbf{y}_t \right) \\ &= -\frac{1}{2} \frac{\alpha \left( \int_{\mathbb{R}^d} p_t^\alpha q_t^{1-\alpha} \|\nabla \log p_t - \nabla \log q_t\|_2^2 d\mathbf{y}_t \right)}{\int_{\mathbb{R}^d} \left( \frac{p_t}{q_t} \right)^\alpha q(\mathbf{y}_t; \boldsymbol{\theta}) d\mathbf{y}_t},\end{aligned}\quad (75)$$

which establishes Eq. (64) in Theorem 3.1.  $\square$

## D.2. Derivation of Variational Rényi Bound

Gibbs variational principle (Polyanskiy & Wu, 2025) shows that

$$\log \mathbb{E}_Q [\exp \{f(X)\}] = \sup_P \mathbb{E}_P [f(X)] - D_{\text{KL}}(P \| Q), \quad (76)$$

where  $f : \mathcal{X} \rightarrow \mathbb{R} \cup \{-\infty\}$  is any measurable function and  $Q$  a probability measure on  $\mathcal{X}$ . The supremum is taken over all  $P$  with  $D_{\text{KL}}(P \| Q) < \infty$ . As a workaround, we get classical ELBO:

$$\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) = \sup_R \left[ \log \mathbb{E}_R \left( \frac{q(\hat{\mathbf{x}}, \mathbf{y}; \boldsymbol{\theta})}{R(\mathbf{x})} \right) \right], \quad (77)$$

where  $R(\cdot)$  is an arbitrary distribution. Amortized variational inference leverages a distribution  $q(\mathbf{y} | \hat{\mathbf{x}}; \boldsymbol{\theta})$  to approximate the intractable posterior  $p(\mathbf{y} | \mathbf{x})$ . This is conceptually related to Bayesian inference, where reasoning about unknown quantities requires posterior distributions that are often intractable. Following variational free-energy approaches with Rényi divergence (Li & Turner, 2016), we approximate the exact posterior  $p(\mathbf{y}_t | \mathbf{x})$  by minimizing  $D_\alpha(p(\mathbf{y} | \mathbf{x}) \| q(\mathbf{y} | \hat{\mathbf{x}}; \boldsymbol{\theta}))$  for  $\alpha > 0$ :

$$D_\alpha(p(\mathbf{y}_t | \mathbf{x}) \| q(\mathbf{y}_t | \hat{\mathbf{x}}; \boldsymbol{\theta})) = \frac{1}{\alpha - 1} \log \int_{\mathcal{Y}} p(\mathbf{y}_t | \mathbf{x}) \left( \frac{p(\mathbf{y}_t | \mathbf{x})}{q(\mathbf{y}_t | \hat{\mathbf{x}}; \boldsymbol{\theta})} \right)^{\alpha - 1} d\mathbf{y}_t \quad (78)$$

$$= -\frac{1}{1 - \alpha} \log \int_{\mathcal{Y}} p(\mathbf{y}_t | \mathbf{x}) \left( \frac{q(\mathbf{y}_t | \hat{\mathbf{x}}; \boldsymbol{\theta})}{p(\mathbf{y}_t | \mathbf{x})} \right)^{1 - \alpha} d\mathbf{y}_t \quad (79)$$

$$= -\frac{1}{1 - \alpha} \log \int_{\mathcal{Y}} p(\mathbf{y}_t | \mathbf{x}) \left( \frac{q(\hat{\mathbf{x}}; \boldsymbol{\theta}) q(\mathbf{y}_t | \hat{\mathbf{x}}; \boldsymbol{\theta})}{p(\mathbf{y}_t | \mathbf{x}) q(\hat{\mathbf{x}}; \boldsymbol{\theta})} \right)^{1 - \alpha} d\mathbf{y}_t \quad (80)$$

$$= -\frac{1}{1 - \alpha} \log \int_{\mathcal{Y}} p(\mathbf{y}_t | \mathbf{x}) \left( \frac{q(\mathbf{y}_t, \hat{\mathbf{x}}; \boldsymbol{\theta})}{p(\mathbf{y}_t | \mathbf{x})} \right)^{1 - \alpha} q(\hat{\mathbf{x}}; \boldsymbol{\theta})^{\alpha - 1} d\mathbf{y}_t \quad (81)$$

$$= -\frac{1}{1 - \alpha} \log q(\hat{\mathbf{x}}; \boldsymbol{\theta})^{\alpha - 1} \int_{\mathcal{Y}} p(\mathbf{y}_t | \mathbf{x}) \left( \frac{q(\mathbf{y}_t, \hat{\mathbf{x}}; \boldsymbol{\theta})}{p(\mathbf{y}_t | \mathbf{x})} \right)^{1 - \alpha} d\mathbf{y}_t \quad (82)$$

$$= \log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) - \frac{1}{1 - \alpha} \log \int_{\mathcal{Y}} p(\mathbf{y}_t | \mathbf{x}) \left( \frac{q(\mathbf{y}_t, \hat{\mathbf{x}}; \boldsymbol{\theta})}{p(\mathbf{y}_t | \mathbf{x})} \right)^{1 - \alpha} d\mathbf{y}_t. \quad (83)$$

Now we consider the equivalent optimization problem  $\max \log q(\hat{\mathbf{x}}; \boldsymbol{\theta})$ :

$$\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) = D_\alpha(p(\mathbf{y}_t | \mathbf{x}) \| q(\mathbf{y}_t | \hat{\mathbf{x}}; \boldsymbol{\theta})) + \frac{1}{1 - \alpha} \log \int_{\mathcal{Y}} p(\mathbf{y}_t | \mathbf{x}) \left( \frac{q(\mathbf{y}_t, \hat{\mathbf{x}}; \boldsymbol{\theta})}{p(\mathbf{y}_t | \mathbf{x})} \right)^{1 - \alpha} d\mathbf{y}_t. \quad (84)$$

Since Rényi divergence is non-negative for positive order  $\alpha$ , we have

$$\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) \geq \frac{1}{1 - \alpha} \log \int_{\mathcal{Y}} p(\mathbf{y}_t | \mathbf{x}) \left( \frac{q(\mathbf{y}_t, \hat{\mathbf{x}}; \boldsymbol{\theta})}{p(\mathbf{y}_t | \mathbf{x})} \right)^{1 - \alpha} d\mathbf{y}_t. \quad (85)$$

In the hierarchical generative models, the model consists of two parts: a forward process forming a conditional joint distribution:  $p(\mathbf{y}_0, \dots, \mathbf{y}_1 | \mathbf{x})$ , and a generative model forming a joint distribution  $q(\mathbf{x}, \mathbf{y}_0, \dots, \mathbf{y}_1)$ . Recall the settings in

Appendix C, we have:

$$-\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) = \frac{1}{\alpha - 1} \log \int_{\mathcal{Y}} p(\mathbf{y}_{0:1} | \mathbf{x}) \left( \frac{q(\mathbf{y}_{0:1}, \hat{\mathbf{x}}; \boldsymbol{\phi}, \boldsymbol{\theta})}{p(\mathbf{y}_{0:1} | \mathbf{x})} \right)^{1-\alpha} d\mathbf{y}_{0:1} - D_{\alpha}(p(\mathbf{y}_{0:1} | \mathbf{x}) \| q(\mathbf{y}_{0:1} | \hat{\mathbf{x}}; \boldsymbol{\theta})) \quad (86)$$

$$\leq \frac{1}{\alpha - 1} \log \int_{\mathcal{Y}} p(\mathbf{y}_{0:1} | \mathbf{x}) \left( \frac{q(\mathbf{y}_{0:1}, \hat{\mathbf{x}}; \boldsymbol{\phi}, \boldsymbol{\theta})}{p(\mathbf{y}_{0:1} | \mathbf{x})} \right)^{1-\alpha} d\mathbf{y}_{0:1} \quad (87)$$

$$= \frac{1}{\alpha - 1} \log \mathbb{E}_{p(\mathbf{y}_{0:1} | \mathbf{x})} \left[ \left( \frac{q(\mathbf{y}_{0:1}, \hat{\mathbf{x}}; \boldsymbol{\theta})}{p(\mathbf{y}_{0:1} | \mathbf{x})} \right)^{1-\alpha} \right] \quad (88)$$

$$= \frac{1}{\alpha - 1} \log \mathbb{E}_{p(\mathbf{y}_{0:1} | \mathbf{x})} \left[ \left( \frac{q(\mathbf{y}_1) q(\hat{\mathbf{x}} | \mathbf{y}_0; \boldsymbol{\theta}) \prod_{i=1}^T q(\mathbf{y}_{s(i)} | \mathbf{y}_{t(i)})}{p(\mathbf{y}_1 | \mathbf{x}) \prod_{i=1}^T p(\mathbf{y}_{s(i)} | \mathbf{y}_{t(i)}, \mathbf{x})} \right)^{1-\alpha} \right] \quad (89)$$

$$= \frac{1}{\alpha - 1} \log \mathbb{E}_{p(\mathbf{y}_{0:1} | \mathbf{x})} \left[ \left( \frac{p(\mathbf{y}_1 | \mathbf{x})}{q(\mathbf{y}_1)} \right)^{\alpha-1} (q(\hat{\mathbf{x}} | \mathbf{y}_0; \boldsymbol{\theta}))^{1-\alpha} \left( \prod_{i=1}^T \frac{q(\mathbf{y}_{s(i)} | \mathbf{y}_{t(i)})}{p(\mathbf{y}_{s(i)} | \mathbf{y}_{t(i)}, \mathbf{x})} \right)^{1-\alpha} \right]. \quad (90)$$

For small enough  $\lambda_{\min}$ ,  $\mathbf{y}_1$  holds almost no information about  $\mathbf{x}$ , such that there exists a distribution  $p(\mathbf{y}_1)$  satisfying  $p(\mathbf{y}_1 | \mathbf{x}) = p(\mathbf{y}_1) \approx q(\mathbf{y}_1)$ . Therefore, the boundary term ratio approaches unity:  $\left( \frac{p(\mathbf{y}_1 | \mathbf{x})}{q(\mathbf{y}_1)} \right)^{\alpha-1} \approx 1$ . Consequently, the bound simplifies to focus on the reconstruction and the diffusion trajectory:

$$-\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) \lesssim \frac{1}{\alpha - 1} \log \mathbb{E}_{p(\mathbf{y}_{0:1} | \mathbf{x})} \left[ \left( \frac{p(\mathbf{y}_1 | \mathbf{x})}{q(\mathbf{y}_1)} \right)^{\alpha-1} (q(\hat{\mathbf{x}} | \mathbf{y}_0; \boldsymbol{\phi}))^{1-\alpha} \left( \prod_{i=1}^T \frac{q(\mathbf{y}_{s(i)} | \mathbf{y}_{t(i)})}{p(\mathbf{y}_{s(i)} | \mathbf{y}_{t(i)}, \mathbf{x})} \right)^{1-\alpha} \right] \quad (91)$$

$$\leq \frac{1}{\alpha - 1} \log \mathbb{E}_{p(\mathbf{y}_{0:1} | \mathbf{x})} \left[ (q(\hat{\mathbf{x}} | \mathbf{y}_0; \boldsymbol{\phi}))^{1-\alpha} \left( \frac{q(\mathbf{y}_{0:1})}{p(\mathbf{y}_{0:1} | \mathbf{x})} \frac{p(\mathbf{y}_1)}{q(\mathbf{y}_1)} \right)^{1-\alpha} \right] \quad (92)$$

$$= \frac{1}{\alpha - 1} \log \mathbb{E}_{p(\mathbf{y}_{0:1} | \mathbf{x})} \left[ (q(\hat{\mathbf{x}} | \mathbf{y}_0; \boldsymbol{\phi}))^{1-\alpha} \left( \frac{q(\mathbf{y}_{0:1})}{p(\mathbf{y}_{0:1} | \mathbf{x})} \right)^{1-\alpha} \right]. \quad (93)$$

In practice, intuitively, given a noisy observation  $\mathbf{y}_t$ , we first make a prediction of the corresponding  $\mathbf{y}_0$  and then use it to obtain a sample  $\hat{\mathbf{x}}$  through the decoder  $q(\hat{\mathbf{x}} | \mathbf{y}_0; \boldsymbol{\phi})$ . By rewriting (28), one can then predict the denoised observation, which is a prediction of  $\mathbf{y}_0$  given  $\mathbf{y}_t$ :

$$f_{\boldsymbol{\theta}}(\mathbf{y}_t, t) := (\mathbf{y}_t - b_t \hat{\boldsymbol{\epsilon}}(\mathbf{y}_t, t; \boldsymbol{\theta})) / a_t. \quad (94)$$

We can then have:

$$-\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) \leq \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})} [-\log q(\hat{\mathbf{x}} | f_{\boldsymbol{\theta}}(\mathbf{y}_t, t); \boldsymbol{\phi})] + D_{\alpha}(p(\mathbf{y}_{0:1} | \mathbf{x}) \| q(\mathbf{y}_{0:1})). \quad (95)$$

Let  $t \in [0, 1]$  and  $\mathbf{y}_t \in \mathbb{R}^d$ . Fix  $\mathbf{x} \in \mathbb{R}^d$ . Consider the variance-preserving (VP) marginal parameterization

$$\mathbf{y}_t = a_t \mathbf{x} + b_t \boldsymbol{\epsilon} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad (96)$$

where

$$a_t^2 = \sigma(\lambda_t), \quad b_t^2 = \sigma(-\lambda_t) = 1 - \sigma(\lambda_t), \quad (97)$$

and  $\lambda \in \mathcal{C}^1([0, 1])$  is strictly decreasing so that  $\dot{\lambda}_t < 0$  and  $a_t, b_t > 0$  for all  $t \in [0, 1]$ . Denote by  $P^x$  the conditional path measure of  $\mathbf{y}_{0:1}$  given  $\mathbf{x}$ , i.e.  $P^x(\cdot) = p(\mathbf{y}_{0:1} | \mathbf{x})$ , and by  $Q$  the (unconditional) model path measure  $Q(\cdot) = q(\mathbf{y}_{0:1})$ .

Throughout, we assume  $P^x$  and  $Q$  are induced by Itô diffusions with the same diffusion coefficient  $g(t)$  and possibly different drifts, ensuring mutual absolute continuity on path space (under standard integrability conditions stated below).

We first identify a forward Itô SDE whose solution admits the above Gaussian marginals. Consider

$$d\mathbf{y}_t = f(t) \mathbf{y}_t dt + g(t) d\mathbf{W}_t \quad \mathbf{y}_0 = \mathbf{x}, \quad (98)$$

where  $\mathbf{W}_t$  is a standard  $d$ -dimensional Brownian motion. This linear SDE admits the explicit solution

$$\mathbf{y}_t = \alpha(t)\mathbf{x} + \int_0^t \alpha(t)\alpha(s)^{-1}g(s)d\mathbf{W}_s \quad \alpha(t) := \exp\left(\int_0^t f(u)du\right). \quad (99)$$

Hence  $(\mathbf{y}_t|\mathbf{x}) \sim \mathcal{N}(\alpha(t)\mathbf{x}; v(t)\mathbf{I})$  with

$$v(t) = \int_0^t \left(\frac{\alpha(t)}{\alpha(s)}\right)^2 g(s)^2 ds. \quad (100)$$

Matching  $\alpha(t) = a_t$  yields

$$f(t) = \frac{\dot{a}_t}{a_t} = \frac{1}{2} \frac{d}{dt} \log a_t^2. \quad (101)$$

In VP form it is standard to write  $f(t) = -\frac{1}{2}\beta(t)$  and  $g(t) = \sqrt{\beta(t)}$  for some  $\beta(t) \geq 0$ . Using  $a_t^2 = \sigma(\lambda_t)$ , we compute

$$\frac{d}{dt} \log a_t^2 = \frac{d}{dt} \log \sigma(\lambda_t) = \sigma(-\lambda_t) \dot{\lambda}_t = b_t^2 \dot{\lambda}_t. \quad (102)$$

For VP diffusions one has  $\frac{d}{dt} \log a_t^2 = -\beta(t)$ , hence

$$\beta(t) = -b_t^2 \dot{\lambda}_t \geq 0 \quad (\dot{\lambda}_t < 0). \quad (103)$$

Therefore, in the log-SNR parameterization,

$$f(t) = -\frac{1}{2}\beta(t) = \frac{1}{2} b_t^2 \dot{\lambda}_t, \quad g(t) = \sqrt{\beta(t)} = \sqrt{-b_t^2 \dot{\lambda}_t}. \quad (104)$$

Let  $\pi_t$  be the time- $t$  marginal density of a diffusion driven by the above forward coefficients  $(f, g)$ . The corresponding reverse-time SDE (from  $t = 1$  to  $t = 0$ ) can be written as

$$d\mathbf{y}_t = \left(f(t)\mathbf{y}_t - g(t)^2 \nabla_{\mathbf{y}} \log \pi_t(\mathbf{y}_t)\right) dt + g(t) d\bar{\mathbf{W}}_t, \quad (105)$$

where  $\bar{\mathbf{W}}_t$  is a reverse-time Brownian motion. Applying this identity to the conditional marginal  $\pi_t(\cdot) = p(\cdot | \mathbf{x})$  and the unconditional marginal  $\pi_t(\cdot) = q_t(\cdot)$  gives:

$$d\mathbf{y}_t = b_p(t, \mathbf{y}_t; \mathbf{x}) dt + g(t) d\bar{\mathbf{W}}_t, \quad b_p(t, \mathbf{y}; \mathbf{x}) := f(t)\mathbf{y} - g(t)^2 \nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t | \mathbf{x}), \quad (106)$$

and

$$d\mathbf{y}_t = b_q(t, \mathbf{y}_t; \theta) dt + g(t) d\bar{\mathbf{W}}_t, \quad b_q(t, \mathbf{y}; \theta) := f(t)\mathbf{y} - g(t)^2 \nabla_{\mathbf{y}_t} \log q_t(\mathbf{y}_t). \quad (107)$$

In our VP setting, the conditional marginal is explicit:

$$p(\mathbf{y}_t | \mathbf{x}) = \mathcal{N}(a_t \mathbf{x}, b_t^2 \mathbf{I}) \Rightarrow \nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t | \mathbf{x}) = -\frac{\mathbf{y}_t - a_t \mathbf{x}}{b_t^2}. \quad (108)$$

Define the (adapted) process

$$\boldsymbol{\theta}_t := \frac{b_p(t, \mathbf{y}_t; \mathbf{x}) - b_q(t, \mathbf{y}_t; \theta)}{g(t)}. \quad (109)$$

Assume the Novikov condition holds under  $Q$ :

$$\mathbb{E}_Q \left[ \exp \left( \frac{1}{2} \int_0^1 \|\boldsymbol{\theta}_t\|^2 dt \right) \right] < \infty. \quad (110)$$

Define the exponential martingale

$$Z_t := \exp \left( \int_0^t \boldsymbol{\theta}_s^\top d\bar{\mathbf{W}}_s - \frac{1}{2} \int_0^t \|\boldsymbol{\theta}_s\|^2 ds \right). \quad (111)$$

Then  $(Z_t)_{t \in [0,1]}$  is a  $Q$ -martingale with  $\mathbb{E}_Q[Z_1] = 1$ . Let  $\tilde{P}^x$  be the measure defined by

$$\frac{d\tilde{P}^x}{dQ} \Big|_{\mathcal{F}_1} = Z_1. \quad (112)$$

By Girsanov's theorem (Song et al., 2021b), the process

$$\bar{\mathbf{W}}_t^{(\tilde{P})} := \bar{\mathbf{W}}_t - \int_0^t \boldsymbol{\theta}_s ds \quad (113)$$

is a Brownian motion under  $\tilde{P}^x$ , and  $\mathbf{y}_t$  satisfies

$$d\mathbf{y}_t = b_p(t, \mathbf{y}_t; \mathbf{x}) dt + g(t) d\bar{\mathbf{W}}_t^{(\tilde{P})}. \quad (114)$$

If, in addition, the terminal marginals differ, i.e.  $\mathbf{y}_1 \sim p(\mathbf{y}_1|\mathbf{x})$  under  $P^x$  and  $\mathbf{y}_1 \sim q(\mathbf{y}_1)$  under  $Q$ , then the Radon–Nikodym derivative on path space is

$$\frac{dP^x}{dQ}(\mathbf{y}_{0:1}) = \frac{p(\mathbf{y}_1|\mathbf{x})}{q(\mathbf{y}_1)} \exp \left( \int_0^1 \boldsymbol{\theta}_t^\top d\bar{\mathbf{W}}_t - \frac{1}{2} \int_0^1 \|\boldsymbol{\theta}_t\|^2 dt \right). \quad (115)$$

In many diffusion-modeling instantiations one sets  $p(\mathbf{y}_1|\mathbf{x}) = q(\mathbf{y}_1)$  (e.g. both standard Gaussian), in which case the terminal density ratio equals one. Let  $L := \frac{dP^x}{dQ}$ . Using the identity

$$D_\alpha(P^x||Q) = \frac{1}{\alpha-1} \log \mathbb{E}_Q[L^\alpha] = \frac{1}{\alpha-1} \log \mathbb{E}_{P^x}[L^{\alpha-1}], \quad (116)$$

and substituting the above expression for  $L$ , standard exponential-martingale calculations yield

$$D_\alpha(P^x||Q) = \frac{1}{\alpha-1} \log \mathbb{E}_{P^x} \left[ \left( \frac{p(\mathbf{y}_1|\mathbf{x})}{q(\mathbf{y}_1)} \right)^{\alpha-1} \exp \left( \frac{\alpha(\alpha-1)}{2} \int_0^1 \frac{\|b_p(t, \mathbf{y}_t; \mathbf{x}) - b_q(t, \mathbf{y}_t; \boldsymbol{\theta})\|^2}{g(t)^2} dt \right) \right]. \quad (117)$$

Under the simplifying assumption  $p(\mathbf{y}_1|\mathbf{x}) = q(\mathbf{y}_1)$ , this reduces to

$$D_\alpha(P^x||Q) = \frac{1}{\alpha-1} \log \mathbb{E}_{P^x} \left[ \exp \left( \frac{\alpha(\alpha-1)}{2} \int_0^1 \frac{\|b_p - b_q\|^2}{g(t)^2} dt \right) \right]. \quad (118)$$

Finally, in the VP log-SNR parameterization specified above,

$$f(t) = \frac{1}{2} b_t^2 \dot{\lambda}_t \quad g(t) = \sqrt{-b_t^2 \dot{\lambda}_t}, \quad (119)$$

so the entire expression is fully determined by  $\lambda_t$  and the model drift (equivalently, the model score).

$$\begin{aligned} D_\alpha(p(\mathbf{y}_{0:1}|\mathbf{x})||q(\mathbf{y}_{0:1})) &= \frac{1}{\alpha-1} \log \mathbb{E}_{P^x} \left[ \exp \left( \frac{\alpha(\alpha-1)}{2} \int_0^1 \frac{\|b_p - b_q\|^2}{g(t)^2} dt \right) \right] \\ &= \frac{1}{\alpha-1} \log \mathbb{E}_{P^x} \left[ \exp \left( \frac{\alpha(\alpha-1)}{2} \int_0^1 \beta(t) \|\nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t|\mathbf{x}) - \hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|^2 dt \right) \right] \\ &= \frac{1}{\alpha-1} \log \mathbb{E}_{P^x} \left[ \exp \left( \frac{\alpha(\alpha-1)}{2} \int_0^1 -\frac{d\lambda}{dt} \|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|^2 dt \right) \right]. \end{aligned} \quad (120)$$

Collecting the results above, we have

$$-\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) \leq \mathbb{E}_{p(\mathbf{y}_0|\mathbf{x})} [-\log q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi})] + \frac{1}{\alpha-1} \log \mathbb{E}_{P^x} \left[ \exp \left( \frac{\alpha(\alpha-1)}{2} \int_0^1 -\frac{d\lambda}{dt} \|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|^2 dt \right) \right]. \quad (121)$$

Consider  $\alpha \leq 1$ , we have, by Jensen's inequality,

$$\begin{aligned}
 -\log q(\hat{\mathbf{x}}; \boldsymbol{\theta}) &\leq \mathbb{E}_{p(\mathbf{y}_0|\mathbf{x})} [-\log q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi})] + \frac{1}{\alpha-1} \log \mathbb{E}_{P^x} \left[ \exp \left( \frac{\alpha(\alpha-1)}{2} \int_0^1 -\frac{d\lambda}{dt} \|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|^2 dt \right) \right] \\
 &\leq \mathbb{E}_{p(\mathbf{y}_0|\mathbf{x})} [-\log q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi})] + \frac{1}{\alpha-1} \mathbb{E}_{P^x} \left[ \log \exp \left( \frac{\alpha(\alpha-1)}{2} \int_0^1 -\frac{d\lambda}{dt} \|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|^2 dt \right) \right] \\
 &\leq \mathbb{E}_{p(\mathbf{y}_0|\mathbf{x})} [-\log q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi})] + \frac{1+\alpha}{4} \mathbb{E}_{t, \mathbf{x}, \boldsymbol{\epsilon}} \left[ \left( -\frac{d\lambda}{dt} \right) \|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|^2 \right].
 \end{aligned} \tag{122}$$

### D.2.1. REDUCTION TO KL DIVERGENCE

Setting  $\varsigma = \alpha - 1$  and defining

$$A(\mathbf{y}_{0:1}) := q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi}) \prod_{i=1}^T \frac{q(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)})}{p(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)}, \mathbf{x})},$$

the bound in (91) can be written as

$$\frac{1}{\alpha-1} \log \mathbb{E}_p [A^{1-\alpha}] = \frac{1}{\varsigma} \log \mathbb{E}_p [e^{-\varsigma \log A}].$$

Taking the limit  $\alpha \rightarrow 1$  (i.e.,  $\varsigma \rightarrow 0$ ) yields

$$\lim_{\alpha \rightarrow 1} \frac{1}{\alpha-1} \log \mathbb{E}_p [A^{1-\alpha}] = -\mathbb{E}_p [\log A],$$

which gives

$$\mathbb{E}_{p(\mathbf{y}_0|\mathbf{x})} [-\log q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi})] + \sum_{i=1}^T \mathbb{E}_{p(\mathbf{y}_{t(i)}|\mathbf{x})} [D_{\text{KL}}(p(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)}, \mathbf{x}) \parallel q(\mathbf{y}_{s(i)}|\mathbf{y}_{t(i)}))] . \tag{123}$$

This recovers the standard variational diffusion objective (Ho et al., 2020; Kingma et al., 2021; Vahdat et al., 2021). Moreover, as  $T \rightarrow \infty$ , the discrete sum converges to the continuous-time form

$$\int_0^1 \mathbb{E}_{p(\mathbf{y}_t|\mathbf{x})} [D_{\text{KL}}(p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x}) \parallel q(\mathbf{y}_s|\mathbf{y}_t))] dt.$$

Furthermore, *reconstruction* term are typically fixed and parameter-free with the case  $\alpha = 1$ . In standard parameterizations (Kingma et al., 2021; Nielsen et al., 2024) make the simplifying assumption that  $q(\hat{\mathbf{x}}|\mathbf{y}_0; \boldsymbol{\phi})$  factorizes over the elements of  $\mathbf{x}$ . Let  $x_i$  be the value of the  $i$ th dimension (i.e., pixel) of  $\mathbf{x}$  and  $z_{0,i}$  the corresponding pixel value of  $\mathbf{y}_0$ :

$$q(\hat{\mathbf{x}}|\mathbf{y}_0) = \prod_i q(x_i|y_{0,i}). \tag{124}$$

In the case of images, the pixel values are assumed to be independent given  $\mathbf{y}_0$  and only dependent on the matching latent component (Kingma et al., 2021). We construct  $q(x_i|y_{0,i})$  from the variational distribution noting that

$$p(\mathbf{x}|\mathbf{y}_0) = \frac{p(\mathbf{y}_0|\mathbf{x})p(\mathbf{x})}{p(\mathbf{y}_0)} \tag{125}$$

and for high enough SNR at  $t = 0$ ,  $p(\mathbf{y}_0|\mathbf{x})$  will be very peaked around  $\mathbf{y}_0 = a_0\mathbf{x}$  as  $b_0^2 \rightarrow 0$ . So we can choose  $q(x_i|y_{0,i}) \propto p(y_{0,i}|x_i)$ , where we normalize over all possible values of  $x_i$ . It means the learning signal stems **solely** from the KL term. While effective, this limits the optimization to a fixed trade-off defined by the KL divergence.

### D.3. Proofs of Corollary 4.1 and Proposition 4.1

#### D.3.1. PROOF OF COROLLARY 4.1

To prove Corollary 4.1, we first require Lemma D.3, the proof of which is provided as followed.

**Lemma D.3** (Score-based Representation of Posterior Covariance (Bao et al., 2022)). *Let  $\nu \sim p(\nu)$  be a latent variable in  $\mathbb{R}^d$ , and let  $\mu$  be the observation generated by the forward diffusion kernel  $p(\mu|\nu) = \mathcal{N}(\mu|a\nu, b^2\mathbf{I})$ . Define the signal-to-noise ratio as  $\text{SNR} = a^2/b^2$ . The expected trace of the conditional covariance of  $\nu$  given  $\mu$  satisfies:*

$$\mathbb{E}_{p(\mu)} \left[ \frac{\text{Tr}(\text{Cov}_{p(\nu|\mu)}[\nu])}{d} \right] = \text{SNR}^{-1} \left( 1 - \frac{b^2}{d} J(\mu) \right), \quad (126)$$

where  $J(\mu) \triangleq \mathbb{E}_{p(\mu)} [\|\nabla_{\mu} \log p(\mu)\|_2^2]$  denotes the Fisher information (or expected score squared norm).

*Proof.* Our derivation proceeds in two steps: first, we establish the relationship between the posterior mean (Tweedie's formula (Efron, 2011; Alain & Bengio, 2014)) and the posterior covariance; second, we apply the trace operator and regularity conditions to obtain the final result.

**Step 1: Relating Posterior Moments via Gradients** Recall that the score function of the marginal distribution  $p(\mu)$  can be expressed as:

$$\nabla_{\mu} \log p(\mu) = \mathbb{E}_{p(\nu|\mu)} [\nabla_{\mu} \log p(\mu|\nu)] = - \frac{\mu - a\mathbb{E}_{p(\nu|\mu)}[\nu]}{b^2}. \quad (127)$$

Rearranging terms yields Tweedie's formula for the posterior mean:

$$\hat{\nu}(\mu) \triangleq \mathbb{E}_{p(\nu|\mu)}[\nu] = \frac{1}{a} (\mu + b^2 \nabla_{\mu} \log p(\mu)). \quad (128)$$

For exponential family distributions (including Gaussians), a known identity links the Jacobian of the posterior mean to the posterior covariance:

$$\nabla_{\mu} \mathbb{E}_{p(\nu|\mu)}[\nu] = \frac{a}{b^2} \text{Cov}_{p(\nu|\mu)}[\nu]. \quad (129)$$

By differentiating Eq. (128) with respect to  $\mu$ , we obtain:

$$\nabla_{\mu} \hat{\nu}(\mu) = \frac{1}{a} (\mathbf{I} + b^2 \nabla_{\mu}^2 \log p(\mu)). \quad (130)$$

Equating the two expressions for the Jacobian allows us to express the posterior covariance analytically in terms of the Hessian of the log-density:

$$\text{Cov}_{p(\nu|\mu)}[\nu] = \frac{b^2}{a^2} (\mathbf{I} + b^2 \nabla_{\mu}^2 \log p(\mu)). \quad (131)$$

**Step 2: Trace and Expectation.** Taking the trace of Eq. (131), we have:

$$\text{Tr}(\text{Cov}_{p(\nu|\mu)}[\nu]) = \frac{b^2}{a^2} (d + b^2 \text{Tr}(\nabla_{\mu}^2 \log p(\mu))). \quad (132)$$

We now take the expectation over  $p(\mu)$ . Under standard regularity conditions (assuming  $p(\mu)$  vanishes at infinity), integration by parts relates the expected Hessian trace to the Fisher information:

$$\mathbb{E}_{p(\mu)} [\text{Tr}(\nabla_{\mu}^2 \log p(\mu))] = -\mathbb{E}_{p(\mu)} [\|\nabla_{\mu} \log p(\mu)\|_2^2] = -J(\mu). \quad (133)$$

Substituting this into the expected trace equation and normalizing by the dimension  $d$ :

$$\mathbb{E}_{p(\mu)} \left[ \frac{\text{Tr}(\text{Cov}_{p(\nu|\mu)}[\nu])}{d} \right] = \frac{b^2}{a^2 d} (d - b^2 J(\mu)) = \frac{b^2}{a^2} \left( 1 - \frac{b^2}{d} J(\mu) \right). \quad (134)$$

Noting that  $\text{SNR}^{-1} = b^2/a^2$ , the proof is complete.  $\square$

**Corollary 4.1** (Optimal Variance under Rényi Divergence). For the diffusion process defined in Appendix C, the variance minimizing (9) is:

$$\sigma_\alpha^{*2} = \sigma_p^2 + \alpha \left( \frac{b_{t|s}^4}{a_{t|s}^2 b_t^2} \right) \left( 1 - \frac{b_t^2}{d} J(\mathbf{y}_t) \right), \quad (135)$$

where  $J(\mathbf{y}_t)$  is the Fisher information (Definition B.2) and  $a_s, a_t, b_s, b_t$  are diffusion coefficients.

*Proof.* We aim to approximate the true posterior transition  $p(\mathbf{y}_s|\mathbf{y}_t)$  with a Gaussian distribution  $q(\mathbf{y}_s|\mathbf{y}_t) = \mathcal{N}(\mathbf{y}_s|\boldsymbol{\mu}_q, \sigma_q^2 \mathbf{I})$ . While the true posterior can be applied to more general Gaussian models, we seek the optimal Gaussian approximation that minimizes the KL divergence,  $D_{\text{KL}}(p(\mathbf{y}_s|\mathbf{y}_t) \| q(\mathbf{y}_s|\mathbf{y}_t))$ . According to (Bao et al., 2022), the optimal parameters correspond to the first and second moments of  $p(\mathbf{y}_s|\mathbf{y}_t)$ :

$$\boldsymbol{\mu}_q^* = \mathbb{E}_{p(\mathbf{y}_s|\mathbf{y}_t)}[\mathbf{y}_s], \quad \sigma_q^{*2} = \mathbb{E}_{p(\mathbf{y}_t)} \left[ \frac{\text{Tr}(\text{Cov}_{p(\mathbf{y}_s|\mathbf{y}_t)}[\mathbf{y}_s])}{d} \right]. \quad (136)$$

Crucially, for Gaussian distributions with matched means, the variance that minimizes the KL divergence also minimizes the Rényi divergence. Consequently, we adopt  $\sigma_q^{*2}$  from (136) to determine the optimal Rényi variance parameter  $\sigma_\alpha^{*2}$ , defined as:

$$\sigma_\alpha^{*2} := (1 - \alpha)\sigma_p^2 + \alpha\sigma_q^{*2}. \quad (137)$$

To derive the analytic form of  $\sigma_q^{*2}$ , we apply the Law of Total Variance, conditioning on the clean data  $\mathbf{x}$ :

$$\begin{aligned} \text{Cov}_{p(\mathbf{y}_s|\mathbf{y}_t)}[\mathbf{y}_s] &= \mathbb{E}_{p(\mathbf{x}|\mathbf{y}_t)} [\text{Cov}_{p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x})}[\mathbf{y}_s]] + \text{Cov}_{p(\mathbf{x}|\mathbf{y}_t)} [\mathbb{E}_{p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x})}[\mathbf{y}_s]] \\ &= \sigma_p^2 \mathbf{I} + \text{Cov}_{p(\mathbf{x}|\mathbf{y}_t)} [\tilde{\boldsymbol{\mu}}_t(\mathbf{y}_t, \mathbf{x})]. \end{aligned} \quad (138)$$

Recall that the posterior mean is linear in  $\mathbf{x}$ , specifically  $\tilde{\boldsymbol{\mu}}_t(\mathbf{y}_t, \mathbf{x}) = C_t \mathbf{x} + \text{const}$ , where the coefficient is given by the diffusion schedule. Substituting the explicit coefficients yields:

$$\text{Cov}_{p(\mathbf{y}_s|\mathbf{y}_t)}[\mathbf{y}_s] = \sigma_p^2 \mathbf{I} + \left( \frac{b_{t|s}^2 a_s}{b_t^2} \right)^2 \text{Cov}_{p(\mathbf{x}|\mathbf{y}_t)}[\mathbf{x}]. \quad (139)$$

We then substitute this into the expression for  $\sigma_q^{*2}$ . Invoking Lemma D.3 to relate the covariance of  $\mathbf{x}$  to the Fisher information  $J(\mathbf{y}_t)$ , we obtain:

$$\begin{aligned} \sigma_q^{*2} &= \sigma_p^2 + (a_s - \sqrt{b_s^2 - \sigma_p^2} \cdot \frac{a_t}{b_t})^2 \mathbb{E}_{p(\mathbf{y}_t)} \left[ \frac{\text{Tr}(\text{Cov}_{p(\mathbf{x}|\mathbf{y}_t)}[\mathbf{x}])}{d} \right] \\ &= \sigma_p^2 + (a_s - \sqrt{b_s^2 - \sigma_p^2} \cdot \frac{a_t}{b_t})^2 \frac{b_t^2}{a_t^2} \left( 1 - \frac{b_t^2}{d} J(\mathbf{y}_t) \right). \end{aligned} \quad (140)$$

Finally, substituting  $\sigma_q^{*2}$  into the definition of  $\sigma_\alpha^{*2}$  yields:

$$\begin{aligned} \sigma_\alpha^{*2} &= (1 - \alpha)\sigma_p^2 + \alpha \left[ \sigma_p^2 + (a_s - \sqrt{b_s^2 - \sigma_p^2} \cdot \frac{a_t}{b_t})^2 \frac{b_t^2}{a_t^2} \left( 1 - \frac{b_t^2}{d} J(\mathbf{y}_t) \right) \right] \\ &= \sigma_p^2 + \alpha \left( \frac{a_s b_t}{a_t} - \sqrt{b_s^2 - \sigma_p^2} \right)^2 \left( 1 - \frac{b_t^2}{d} J(\mathbf{y}_t) \right) \\ &= \sigma_p^2 + \alpha \left( \frac{b_{t|s}^4}{a_{t|s}^2 b_t^2} \right) \left( 1 - \frac{b_t^2}{d} J(\mathbf{y}_t) \right), \end{aligned}$$

where the final step follows from the algebraic identities of the diffusion coefficients  $a_{t|s}$  and  $b_{t|s}$ .  $\square$

## D.3.2. PROOF OF PROPOSITION 4.1

Let  $\mathbf{y}_t \in \mathbb{R}^d$  be a random vector (e.g., the noisy data at timestep  $t$ ) with a smooth probability density function  $p(\mathbf{y}_t)$ . Let  $\mathcal{H}(\mathbf{y}_t)$  denote its differential entropy and  $J(\mathbf{y}_t)$  its Fisher information.

We assert that the following inequality holds:

$$\frac{J(\mathbf{y}_t)}{d} \geq 2\pi e \cdot e^{-\frac{2}{d}\mathcal{H}(\mathbf{y}_t)} \quad (141)$$

Equivalently, in the negative form:

$$-\frac{J(\mathbf{y}_t)}{d} \leq -2\pi e \cdot e^{-\frac{2}{d}\mathcal{H}(\mathbf{y}_t)} \quad (142)$$

This result is a direct consequence of Stam's Inequality (Stam, 1959), which states that for a fixed differential entropy, the Gaussian distribution minimizes the Fisher information. The proof proceeds by constructing an "entropy-matched" Gaussian distribution.

Consider an auxiliary  $d$ -dimensional spherical Gaussian random vector  $\mathbf{z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ . We choose the variance  $\sigma^2$  such that the entropy of  $\mathbf{z}$  matches the entropy of  $\mathbf{y}_t$ :

$$\mathcal{H}(\mathbf{z}) = \mathcal{H}(\mathbf{y}_t) \quad (143)$$

The differential entropy of the  $d$ -dimensional spherical Gaussian  $\mathbf{z}$  is given by:

$$\mathcal{H}(\mathbf{z}) = \frac{d}{2} \ln(2\pi e \sigma^2) \quad (144)$$

By inverting this equation, we solve for the variance  $\sigma^2$  (often referred to as the entropy power):

$$\sigma^2 = \frac{1}{2\pi e} e^{\frac{2}{d}\mathcal{H}(\mathbf{z})} \quad (145)$$

The Fisher information is defined as  $J(\mathbf{z}) = \mathbb{E}[\|\nabla \log p(\mathbf{z})\|_2^2]$ . For  $\mathbf{z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ , the score function is  $\nabla \log p(\mathbf{z}) = -\frac{\mathbf{z}}{\sigma^2}$ . Thus, the Fisher information is:

$$J(\mathbf{z}) = \mathbb{E}\left[\left\|-\frac{\mathbf{z}}{\sigma^2}\right\|^2\right] = \frac{1}{\sigma^4} \mathbb{E}[\|\mathbf{z}\|^2] = \frac{d\sigma^2}{\sigma^4} = \frac{d}{\sigma^2} \quad (146)$$

Substituting Eq. (145) into the expression for  $J(\mathbf{z})$ , and recalling that  $\mathcal{H}(\mathbf{z}) = \mathcal{H}(\mathbf{y}_t)$ , we obtain:

$$J(\mathbf{z}) = \frac{d}{\frac{1}{2\pi e} e^{\frac{2}{d}\mathcal{H}(\mathbf{y}_t)}} = d \cdot 2\pi e \cdot e^{-\frac{2}{d}\mathcal{H}(\mathbf{y}_t)} \quad (147)$$

Stam's inequality asserts that among all distributions with a given entropy, the Gaussian distribution has the lowest Fisher information. Therefore:

$$J(\mathbf{y}_t) \geq J(\mathbf{z}) \quad (148)$$

Substituting Eq. (147) into this inequality yields:

$$J(\mathbf{y}_t) \geq d \cdot 2\pi e \cdot e^{-\frac{2}{d}\mathcal{H}(\mathbf{y}_t)} \quad (149)$$

Dividing by and rearranging gives the final result:

$$-\frac{J(\mathbf{y}_t)}{d} \leq -2\pi e \cdot e^{-\frac{2}{d}\mathcal{H}(\mathbf{y}_t)} \quad (150)$$

This inequality establishes a fundamental lower bound on the Fisher information  $J(\mathbf{y}_t)$ , which corresponds to the score matching loss, for the variable  $\mathbf{y}_t$ . The equality holds if and only if  $\mathbf{y}_t$  follows a Gaussian distribution. In the context of diffusion models, as the timestep  $t$  increases and the signal becomes increasingly dominated by Gaussian noise, the distribution  $p(\mathbf{y}_t)$  converges towards a Gaussian, thereby driving  $J(\mathbf{y}_t)$  to approach this theoretical limit.

Substituting this bound into the expression for  $\sigma_q^{*2}$  yields the following upper bound:

$$\sigma_q^{*2} \leq \sigma_p^2 + \left( a_s - \sqrt{b_s^2 - \sigma_p^2} \cdot \frac{a_t}{b_t} \right)^2 \frac{b_t^2}{a_t^2} \left( 1 - 2\pi e b_t^2 e^{-\frac{2}{d} \mathcal{H}(\mathbf{y}_t)} \right). \quad (151)$$

By defining the entropy power variance as  $\sigma_e^2(\mathbf{y}_t) \triangleq \frac{1}{2\pi e} e^{\frac{2}{d} \mathcal{H}(\mathbf{y}_t)}$  and simplifying the squared term, we obtain the final bounds for  $\sigma_q^{*2}$ :

$$\sigma_p^2 \leq \sigma_q^{*2} \leq \sigma_p^2 + \left( \frac{a_s b_t}{a_t} - \sqrt{b_s^2 - \sigma_p^2} \right)^2 \left( 1 - \frac{b_t^2}{\sigma_e^2(\mathbf{y}_t)} \right), \quad (152)$$

where the lower bound  $\sigma_p^2$  follows from the non-negativity of the second term (since  $\sigma_e^2(\mathbf{y}_t) \geq b_t^2$ ).

## E. Equivalence of different predictors and matching objectives

In the case of finite  $T$ , using  $s$  and  $t$  as shorthands for  $s(i) = (i-1)/T$  and  $t(i) = i/T$ , the Rényi diffusion divergence loss is:

$$\mathcal{L}_\alpha^T(\mathbf{x}; \boldsymbol{\theta}) := \sum_{i=1}^T \mathbb{E}_{p(\mathbf{y}_t|\mathbf{x})} [D_\alpha(p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x}) \| q(\mathbf{y}_s|\mathbf{y}_t; \boldsymbol{\theta}))]. \quad (153)$$

We will first derive an expression of  $D_\alpha(p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x}) \| q(\mathbf{y}_s|\mathbf{y}_t; \boldsymbol{\theta}))$ . Recall the reverse-time diffusion process:

$$p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x}) = \mathcal{N} \left( \frac{a_{t|s} b_s^2}{b_t^2} \mathbf{y}_t + \frac{b_{t|s}^2 a_s}{b_t^2} \mathbf{x}, \sigma_p^2 \mathbf{I} \right), \quad (154)$$

and learned reverse kernel:

$$q(\mathbf{y}_s|\mathbf{y}_t; \boldsymbol{\theta}) = \mathcal{N} \left( \frac{a_{t|s} b_s^2}{b_t^2} \mathbf{y}_t + \frac{b_{t|s}^2 a_s}{b_t^2} \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta}), \sigma_q^2 \mathbf{I} \right). \quad (155)$$

Since  $p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x})$  and  $q(\mathbf{y}_s|\mathbf{y}_t; \boldsymbol{\theta})$  are Gaussians, their Rényi divergence is available in closed form as a function of their means and variance:

$$\begin{aligned} D_\alpha(p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x}) \| q(\mathbf{y}_s|\mathbf{y}_t; \boldsymbol{\theta})) &= \frac{\alpha \|\boldsymbol{\mu}_p - \boldsymbol{\mu}_q\|_2^2}{2\sigma_\alpha^2} + \frac{d}{2(1-\alpha)} \log \frac{\sigma_\alpha^2}{\sigma_p^{2(1-\alpha)} \sigma_q^{2\alpha}} \\ &= \frac{\alpha \left\| \frac{a_{t|s} b_s^2}{b_t^2} \mathbf{y}_t + \frac{b_{t|s}^2 a_s}{b_t^2} \mathbf{x} - \frac{a_{t|s} b_s^2}{b_t^2} \mathbf{y}_t + \frac{b_{t|s}^2 a_s}{b_t^2} \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta}) \right\|_2^2}{2\sigma_\alpha^2} + \frac{d}{2(1-\alpha)} \log \frac{\sigma_\alpha^2}{\sigma_p^{2(1-\alpha)} \sigma_q^{2\alpha}} \\ &= \frac{\alpha \left\| \frac{b_{t|s}^2 a_s}{b_t^2} \mathbf{x} - \frac{b_{t|s}^2 a_s}{b_t^2} \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta}) \right\|_2^2}{2\sigma_\alpha^2} + \frac{d}{2(1-\alpha)} \log \frac{\sigma_\alpha^2}{\sigma_p^{2(1-\alpha)} \sigma_q^{2\alpha}} \\ &\stackrel{(i)}{=} \frac{\alpha \left\| \frac{b_{t|s}^2 a_s}{b_t^2} \mathbf{x} - \frac{b_{t|s}^2 a_s}{b_t^2} \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta}) \right\|_2^2}{2\sigma_p^2} \\ &= \frac{\alpha}{2} (\text{SNR}(s) - \text{SNR}(t)) \|\mathbf{x} - \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2, \end{aligned} \quad (156)$$

where in (i) we have used the  $\sigma_p^2 = \sigma_q^2$ . To calculate the limit of the diffusion loss as  $T \rightarrow \infty$ , we express  $\mathcal{L}$  as a function of

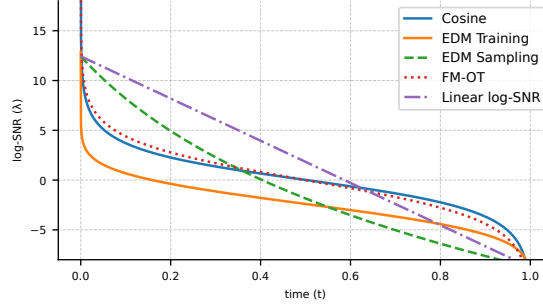


Figure 5. Commonly used log-SNR schedules: cosine (Nichol & Dhariwal, 2021), EDM (Karras et al., 2022), Flow Matching with OT (Lipman et al., 2023) and linear (Nielsen et al., 2024).

$\tau = 1/T$ :

$$\begin{aligned}
 \lim_{T \rightarrow \infty} \sum_{i=1}^T \mathbb{E}_{p(\mathbf{y}_t|\mathbf{x})} [D_\alpha(p(\mathbf{y}_s|\mathbf{y}_t, \mathbf{x}) \| q(\mathbf{y}_s|\mathbf{y}_t; \boldsymbol{\theta}))] &= \lim_{\tau \rightarrow 0} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}\{1, T\}} \left[ \frac{\alpha (\text{SNR}(t - \tau) - \text{SNR}(t))}{2\tau} \|\mathbf{x} - \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2 \right] \\
 &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \int_0^1 \left[ -\frac{\alpha}{2} \text{SNR}'(t) \|\mathbf{x} - \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2 \right] dt \quad (157) \\
 &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}(0, 1)} \left[ -\frac{\alpha}{2} \text{SNR}'(t) \|\mathbf{x} - \hat{\mathbf{x}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2 \right] \quad (158) \\
 &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}(0, 1)} \left[ -\frac{\alpha}{2} \frac{d\lambda}{dt} \|\epsilon - \hat{\epsilon}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2 \right] \quad (159) \\
 &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}(0, 1)} \left[ -\frac{\alpha}{2b_t^2} \frac{d\lambda}{dt} \|\nabla \log p(\mathbf{y}_t|\mathbf{x}) - \hat{\mathbf{s}}(\mathbf{y}_t, t; \boldsymbol{\theta})\|_2^2 \right]. \quad (160)
 \end{aligned}$$

In summary, under different parameterizations of the score network, the  $\epsilon$ -prediction loss admits several equivalent forms:

$$\|\epsilon - \hat{\epsilon}(\boldsymbol{\theta})\|_2^2 = \text{SNR}(t) \|\mathbf{x} - \hat{\mathbf{x}}(\boldsymbol{\theta})\|_2^2 = b_t^2 \|\nabla_{\mathbf{y}_t} \log p(\mathbf{y}_t|\mathbf{x}) - \hat{\mathbf{s}}(\boldsymbol{\theta})\|_2^2 = (e^{-\lambda(t)} + 1)^{-1} \|\mathbf{v} - \hat{\mathbf{v}}(\boldsymbol{\theta})\|_2^2. \quad (161)$$

These equivalences highlight that different prediction targets, namely  $\epsilon$ ,  $\mathbf{x}$ , the score function, or the  $\mathbf{v}$ -parameterization, are related through simple time-dependent rescalings. We refer to (Kingma & Gao, 2023; Zheng et al., 2023c) for a comprehensive discussion of these parameterizations and their practical implications.

## F. Noise schedules

During model training, we sample time  $t$  uniformly,  $t \sim \mathcal{U}(0, 1)$ , and then compute  $\lambda = f_\lambda(t)$ . This induces a distribution over noise levels  $p(\lambda)$ , whose cumulative density function (CDF) is given by

$$F(\lambda) = 1 - f_\lambda^{-1}(\lambda). \quad (162)$$

For  $\lambda \in [\lambda_{\min}, \lambda_{\max}]$ , the corresponding probability density function (PDF) is obtained by differentiating the CDF:

$$p(\lambda_t) = -\frac{d}{d\lambda} f_\lambda^{-1}(\lambda) = -\frac{dt}{d\lambda} = -\frac{1}{f'_\lambda(t)}. \quad (163)$$

Outside the interval  $[\lambda_{\min}, \lambda_{\max}]$ , the probability density is zero. In Fig. 5, we summarize several commonly used noise schedules, including the cosine schedule (Nichol & Dhariwal, 2021), the shifted cosine schedule (Hoogeboom et al., 2023), and the EDM schedule (Karras et al., 2022). We do not include the linear schedule proposed in (Ho et al., 2020),

$$f_\lambda(t) = -\log(e^{t^2} - 1), \quad (164)$$

as it has largely fallen out of use.

## G. Experimental Settings

In this section, we provide details of our experiment settings.

**Training procedure.** Across all experiments, the diffusion time variable  $t$  is sampled using the low-discrepancy time sampler introduced in (Kingma et al., 2021), rather than uniform sampling, which has been shown to reduce the variance of the diffusion loss estimator and accelerate optimization. Unless otherwise specified, we follow the default training configuration of (Kingma et al., 2021).

All models are optimized using Adam (Kingma & Ba, 2015) with a learning rate of  $2 \times 10^{-4}$ , exponential decay rates  $\beta_1 = 0.9$  and  $\beta_2 = 0.99$ , and decoupled weight decay with coefficient 0.01 followed (Zheng et al., 2023c). Gradient norms are clipped to 1 for training stability. For evaluation, we maintain an exponential moving average (EMA) of model parameters with a decay rate of 0.9999.

**Adaptive noise scheduling.** We implemented an adaptive noise schedule  $\rho(\lambda_t)$  taken directly from (Kingma & Gao, 2023), where:

$$\rho(\lambda_t) \propto \mathbb{E}_{\mathbf{x}, \epsilon} [w(\lambda) \|\epsilon - \hat{\epsilon}(\mathbf{y}_t, \lambda; \theta)\|_2^2]. \quad (165)$$

For experiments involving adaptive noise schedules, the continuous noise-level range  $[\lambda_{\min}, \lambda_{\max}]$  is discretized into 100 evenly spaced bins. During training, we maintain an exponential moving average of the weighted denoising error  $w(\lambda) \|\epsilon - \hat{\epsilon}(\mathbf{y}_t, \lambda; \theta)\|_2^2$  with a decay rate 0.999. This running statistic is used solely to estimate relative loss scales across noise levels and does not introduce additional trainable parameters.

**Datasets** We perform experiments on CIFAR-10 and ImageNet datasets. CIFAR-10 contains 50,000 training and 10,000 test images. The ImageNet variant includes 1,281,149 training and 49,999 test images. Among the two known versions of ImageNet32, we adopt the newer, anti-aliased version, which facilitates likelihood training and remains publicly available. The older version used in (Song et al., 2021b; Kingma et al., 2021) is no longer accessible. Furthermore, it is notable that ImageNet contains some personal sensitive information and may cause privacy concern (Song et al., 2021b).

**CIFAR-10 and ImageNet-32.** For CIFAR-10 and ImageNet-32, we employ U-Net-based diffusion models parameterized using  $\epsilon$ -prediction and  $v$ -prediction, following the architectures introduced in (Kingma et al., 2021; Zheng et al., 2023c). These architectures are specifically designed for likelihood-based training, incorporating modifications such as the removal of internal downsampling/upsampling layers and the use of Fourier features to better capture fine-scale details. Consistent with (Kingma et al., 2021), we use a U-Net of depth 32 with 128 channels for CIFAR-10, and a U-Net of the same depth with 256 channels for ImageNet-32. A dropout rate of 0.1 is applied in intermediate layers for regularization. For other hyperparameters, we use fixed start and end times which satisfy  $\lambda_{\min} = -13.3$  and  $\lambda_{\max} = 5.0$ , which is the default setting in (Kingma et al., 2021). All our training processes are conducted on 8 GPU cards of NVIDIA L40 for ImageNet-32 and CIFAR-10.

**ImageNet-64.** For class-conditional generation on ImageNet-64, we adopt the ADM U-Net architecture of (Dhariwal & Nichol, 2021), with a dropout rate of 0.1 and no data augmentation. Training largely follows the configuration of (Karras et al., 2022), using a learning rate of  $10^{-4}$ , an exponential moving average over 50 million images, and a learning-rate warmup over 10 million images. Models are trained on NVIDIA H20 GPUs with a total batch size of 2048. Checkpoints are saved periodically and evaluated, and we report results from the best-performing checkpoint. For class-conditional generation on higher resolution ImageNet, we heavily followed the setting of simple diffusion (Hooeboom et al., 2023). Specifically, we used their U-ViT architecture, and followed their learning rate and EMA schedules.

For the forward diffusion process, we set  $\lambda_{\min} = -15$  and  $\lambda_{\max} = 15$ . During sampling, we fix the noise schedule to the cosine schedule for the DDPM sampler and to the standard EDM sampling schedule for the EDM sampler. All EDM sampling hyperparameters are taken directly from (Karras et al., 2022) without modification. Both DDPM and EDM samplers use 250 sampling steps.

## H. Sample Quality and FID

To quantitatively measure these aspects, numerous metrics have been proposed, including IS, FID, Kernel Inception Distance (Bińkowski et al., 2018), and Precision/Recall (Sajjadi et al., 2018). Among these metrics, the FID, defined

Table 7. Sample quality (FID &amp; IS) and number of function evaluations (NFE) on CIFAR-10 and ImageNet-512.

| Unconditional CIFAR-10                      |                   |                     |                      | Class-Conditional ImageNet 512x512                     |                   |                     |                      |
|---------------------------------------------|-------------------|---------------------|----------------------|--------------------------------------------------------|-------------------|---------------------|----------------------|
| METHOD                                      | IS ( $\uparrow$ ) | NFE( $\downarrow$ ) | FID ( $\downarrow$ ) | METHOD                                                 | IS ( $\uparrow$ ) | NFE( $\downarrow$ ) | FID ( $\downarrow$ ) |
| <b>Diffusion models &amp; Fast Samplers</b> |                   |                     |                      | <b>Diffusion models</b>                                |                   |                     |                      |
| Score SDE (Song et al., 2021c)              | 9.89              | 2000                | 2.20                 | ADM-G (Dhariwal & Nichol, 2021)                        | 172.71            | 250                 | 7.72                 |
| EDM (Karras et al., 2022)                   | -                 | 35                  | 1.97                 | RIN (Jabri et al., 2023)                               | 216.00            | 1000                | 3.95                 |
| Flow Matching (Lipman et al., 2023)         | -                 | 142                 | 6.35                 | ADM-U (Dhariwal & Nichol, 2021)                        | 221.72            | 250                 | 3.85                 |
| OT-CFM (Tong et al., 2024)                  | -                 | 1000                | 3.57                 | U-ViT-H/4 (Bao et al., 2023)                           | 263.79            | 250                 | 4.05                 |
| DPM-Solver (Lu et al., 2022b)               | -                 | 10                  | 4.70                 | VDM++ (Kingma & Gao, 2023)                             | 278.10            | 250                 | 2.65                 |
| DPM-Solver++ (Lu et al., 2025)              | -                 | 10                  | 2.91                 | DiffiT (Hatamizadeh et al., 2024)                      | 252.12            | 250                 | 2.67                 |
| DPM-Solver-v3 (Zheng et al., 2023b)         | -                 | 10                  | 2.51                 | EDM2-XXL (interval) (Kynkäänniemi et al., 2024)        | -                 | 63                  | <b>1.40</b>          |
| <b>Joint Training</b>                       |                   |                     |                      | <b>GANs &amp; Masked Models</b>                        |                   |                     |                      |
| Diffusion GAN (Xiao et al., 2022)           | 9.63              | 4                   | 3.75                 | StyleGAN-XL (Sauer et al., 2022)                       | 267.75            | 1                   | 2.40                 |
| Diffusion StyleGAN (Wang et al., 2023)      | 9.94              | 1                   | 3.19                 | MAGViT-v2 (Yu et al., 2024)                            | <b>324.30</b>     | 64                  | 1.91                 |
| R3GAN (Huang et al., 2024)                  | -                 | 1                   | 1.96                 | MAR (Li et al., 2024b)                                 | 279.90            | 64                  | 1.73                 |
| GMem (Tang et al., 2024)                    | -                 | 50                  | <b>1.22</b>          | VAR-d36-s (Tian et al., 2024)                          | 303.20            | 10                  | 2.63                 |
| CTM (Kim et al., 2024a)                     | -                 | 1                   | 1.87                 | BigGAN (Brock et al., 2019)                            | 177.90            | 1                   | 8.43                 |
| DDO (Zheng et al., 2025a)                   | -                 | 35                  | 1.38                 | VQGAN (Esser et al., 2021)                             | 66.80             | 1024                | 26.52                |
| DMD (Yin et al., 2024)                      | -                 | 1                   | 3.77                 | PaGoDA (Kim et al., 2024b)                             | 251.31            | 1                   | 1.80                 |
| SiD (Zhou et al., 2024)                     | 10.02             | 1                   | 1.92                 | MaskDiT (Zheng et al., 2023a)                          | -                 | 79                  | 2.50                 |
| GDD-I (Zheng & Yang, 2024)                  | 10.11             | 1                   | 1.54                 |                                                        |                   |                     |                      |
| StyleGAN2-ADA + SMaRt (Xia et al., 2024)    | <b>10.22</b>      | 1                   | 2.06                 |                                                        |                   |                     |                      |
| <b>Diffusion Distillation</b>               |                   |                     |                      | <b>Diffusion Models</b>                                |                   |                     |                      |
| DFNO (LPIPS) (Zheng et al., 2023a)          | -                 | 1                   | 3.78                 | SimDiff (Hooeboom et al., 2023)                        | 248.70            | 512                 | 3.02                 |
| 2-Rectified Flow (Liu et al., 2022)         | 9.60              | 127                 | 2.58                 | DiT-XL/2 (Peebles & Xie, 2023)                         | 240.80            | 250                 | 3.04                 |
| PID (LPIPS) (Tee et al., 2024)              | 9.13              | 1                   | 3.92                 | DiMR-XL/3R (Liu et al., 2024)                          | 289.80            | 250                 | 2.89                 |
| Consistency-FM (Yang et al., 2024)          | 8.70              | 2                   | 5.34                 | SimDiff-2 (Hooeboom et al., 2025)                      | -                 | 512                 | 1.48                 |
| PD (Salimans & Ho, 2022)                    | 8.69              | 1                   | 8.34                 | DiS-H/2 (Fei et al., 2024a)                            | 272.33            | 250                 | 2.88                 |
| TRACT (Berthelot et al., 2023)              | -                 | 2                   | 5.88                 | DRWKV-H/2 (Fei et al., 2024b)                          | 265.20            | 250                 | 2.95                 |
| CD (LPIPS) (Song et al., 2023)              | -                 | 2                   | 2.24                 | LDM-4-G (Rombach et al., 2022)                         | 247.67            | 1                   | 3.60                 |
| Diff-Instruct (Luo et al., 2023)            | 9.89              | 1                   | 2.71                 | EDM2-XXL (Karras et al., 2024)                         | 309.80            | 63                  | 1.73                 |
| <b>Consistency Training</b>                 |                   |                     |                      | <b>Consistency Model</b>                               |                   |                     |                      |
| iCT (Song & Dhariwal, 2024)                 | 9.54              | 2                   | 2.46                 | sCD-XXL (Lu & Song, 2025)                              | -                 | 1                   | 2.28                 |
| iCT-deep (Song & Dhariwal, 2024)            | 9.76              | 2                   | 2.24                 | -                                                      | -                 | 2                   | 1.88                 |
| ECM (Geng et al., 2025)                     | -                 | 2                   | 2.11                 | sCT-XXL (Lu & Song, 2025)                              | -                 | 1                   | 4.29                 |
| sCT (Lu & Song, 2025)                       | -                 | 2                   | 2.06                 | -                                                      | -                 | 2                   | 3.76                 |
| <b>Our Diffusion Model</b>                  |                   |                     |                      | <b>Our Diffusion Model</b>                             |                   |                     |                      |
| RDM (ours), EDM weighting                   | 9.73              | 250                 | 2.13                 | RDM (ours), sigmoidal weighting (train), with guidance | 279.58 $\pm$ 4.7  | 250                 | 2.77                 |
| RDM (ours), EDM-monotonic weighting         | 9.92              | 250                 | 2.16                 | RDM (ours), sigmoidal weighting (test), with guidance  | -                 | 250                 | 4.17                 |
| RDM (ours), sigmoidal weighting             | 9.89              | 250                 | 2.07                 | RDM (ours), without guidance                           | 237.17 $\pm$ 3.8  | 250                 | 2.89                 |

using the sample means and covariances of real and generated embeddings  $(\mu_r, \Sigma_r)$  and  $(\mu_g, \Sigma_g)$  as  $\|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$ , remains the dominant metric for quantifying sample quality. By measuring the distance between real and generated embeddings, FID addresses a central limitation of IS, which does not compare against real data. Accordingly, we include IS only for completeness and focus our evaluation on FID, leaving other metrics such as Precision/Recall to future work. Note that our primary focus is density estimation and probabilistic modeling rather than sample quality. The visual appeal of generated images (as measured by e.g., FID) correlates imperfectly with log-likelihood.

Despite focusing primarily on improving log-likelihood estimation, our experimental results demonstrate that RDM maintains strong and competitive FID performance across various image datasets. We report FID for completeness, while leaving dedicated sample-quality optimization to future work.

## I. Geometry Intuition

Notably, in high-dimensional models, learning takes place in a parameter space, which is generally not Euclidean but Riemannian (Amari, 2016). This calls for incorporating the underlying Riemannian structure into the design of learning objectives. In the contexts of differential geometry, where Fisher information induces a local Riemannian metric  $\mathbf{G}$ , both the objective in (3) and the metric  $\mathbf{G}$  originate from the same domain: the *score* function over data space (a.k.a., the ambient space). Moreover, both real and generated data can be regarded as low-dimensional manifolds embedded in high-dimensional pixel space (Arjovsky & Bottou, 2017; Bortoli, 2022), which indicates that, in general, the two manifolds either do not overlap or intersect only in a transversal manner. This perspective implies that the set difference between the generated and real data manifolds typically has positive Lebesgue measure, indicating that the generated distribution fails to fully cover the data manifold.

However, there is a significant challenge if we only consider the original (unperturbed) data space. The score will be undefined in the ambient space, and score matching will fail to provide a consistent score estimator. It is proposed to add

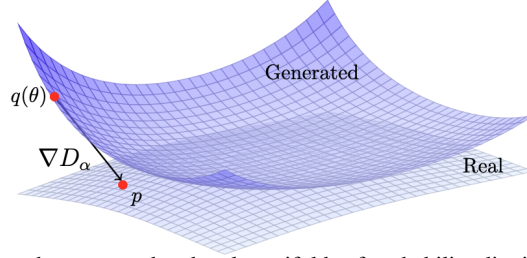


Figure 6. Blue and purple surfaces denote the generated and real manifolds of probability distributions, respectively. The proposed Rényi score matching provides learning trajectory guidance, urging the hypothesis manifold to move towards the real data manifold. In this case, learning loss (Rényi divergence) regains to exert effective guidance aiding the generative distribution to converge to the real one.

Table 8. Lossless compression performance on CIFAR-10 in bits/dim.

| Model                          | Compression Rate (bits/dim) |
|--------------------------------|-----------------------------|
| FLIF (Sneyers & Wuille, 2016)  | 4.14                        |
| LBB (Ho et al., 2019)          | 3.12                        |
| IDF (Hoogetboom et al., 2019)  | 3.26                        |
| VDM (Kingma et al., 2021)      | 2.72                        |
| ARDM (Hoogetboom et al., 2022) | 2.71                        |
| W-PCDM (Li et al., 2024a)      | <b>2.37</b>                 |
| <b>Ours</b>                    | 2.41                        |

random noise to ensure the resulting distribution does not collapse to a low dimensional manifold, which, in fact, improves score estimation. Motivated by this, we extend the analysis in (Xia et al., 2024) to a much more general setting, to study the behavior of information geometry with noise.

The skew divergence  $\mathcal{J}_\alpha$  quantifies the dissimilarities between probability distributions within a Riemannian manifold, measuring how the information structures of  $p$  and  $q(\theta)$  deviate across the space. The intuitive nature of Rényi divergence is evident in the preceding theorems, as both sides of the equation capture the discrepancies between the true distribution and model-induced one due to noise perturbations. Notably, when  $p$  and  $q(\theta)$  coincide, the divergence vanishes, meaning that all generated samples lie exactly on the data manifold, thereby mitigating the gradient vanishing problem simultaneously due to the intrinsic property that the real and generated data manifolds are almost surely non-coincident. Moreover, the negative gradient with respect to variance highlights how the divergence decreases under diffusion. The rate of change of Rényi divergence encodes local geometric properties, where the relative Rényi-Fisher information encodes this structure, capturing how noise influences the geometry of learning and provides gradient for sampler persistently to guide  $\hat{x}(\theta)$  to lie on data manifold.

To build geometric intuition, note that a divergence measures the degree of separation between two distributions but generally lacks symmetry, and hence is not a true distance. This implies that the generated distribution typically differs from the data manifold in a measurable way. From a geometric standpoint, the true distribution  $p$  induces a natural information manifold, and the model distribution  $q(\theta)$  seeks to approximate it. The mismatch in their score functions, interpreted as gradient vector fields of the perturbed data distribution across all noise levels, can be viewed as a local notion of discrepancy in the underlying Riemannian space. Unlike KL divergence, the Rényi divergence is *skew symmetric* for  $\alpha \notin \{0, 1\}$  (van Erven & Harremos, 2014), capturing how one distribution’s flow deviates from another. Intuitively, Rényi score matching encourages off-manifold points to move closer to the data manifold, narrowing the gap between the generated samples and real data.

## J. Additional Experiments

**Lossless Progressive Coding** As shown in prior work (Ho et al., 2020; Kingma et al., 2021), likelihood-based generative models can be viewed as latent-variable models for neural lossless compression. We adopt this perspective and implement a Bits-Back Coding scheme (Ho et al., 2019) using our proposed model as the latent component. On CIFAR-10, our method achieves shorter average code lengths, measured in bits per dimension, compared to several strong baselines (see Table 8). We leave this avenue of research for further work.

## K. Additional Samples



Figure 7. Random samples from *Rényi Diffusion Model* trained on CIFAR-10. Left: DDPM (Ho et al., 2020) sampler. Right: EDM (Karras et al., 2022) sampler.

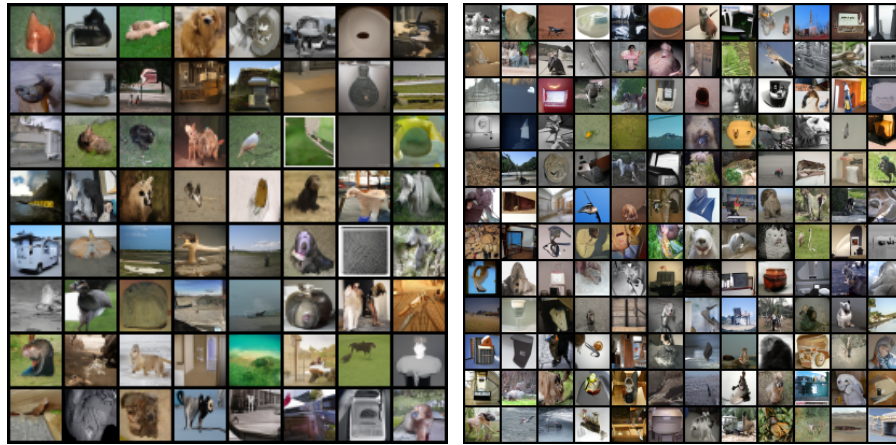


Figure 8. Random samples from *Rényi Diffusion Model* trained on ImageNet-32. Left: DDPM (Ho et al., 2020) sampler. Right: EDM (Karras et al., 2022) sampler.